

Stochastic Processes

MATH 447

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CONTENTS

I	Discrete-Time Markov Chains	3
	Introduction	3
	<i>Time-Homogeneous Markov Chains</i>	
	<i>Multi-Step Transition Probabilities</i>	
	Stationary and Limiting Distributions	9
	<i>Periodicity of States</i>	
	<i>Finding Stationary Distributions</i>	
	<i>Transience and Recurrence</i>	
	<i>Canonical Decompositions</i>	
	<i>Proof of Fundamental Theorem of Markov Chains</i>	
	Hitting Times and Absorbing States	20
	Reversibility	23
	<i>Metropolis-Hastings</i>	
II	Branching Processes	28
	<i>Generating Functions</i>	
III	Poisson Processes	33
	<i>Examples, Intuition, and First Definitions</i>	
	<i>One Dimensional Processes</i>	
	<i>Variable-Rate Processes</i>	
	<i>Thinning and Superposition</i>	
IV	Continuous-Time Markov Chains	37
	Stationary and Limiting Distributions	44
	<i>Fundamental Theorem of Continuous-Time Markov Chains</i>	
	Yule Processes	47
	<i>Birth-and-Death Processes</i>	
	A Final Example	50

Definitions

Theorems

We assume working knowledge of probability and no knowledge of measure theory (though a grasp of analysis is essential). See these [MATH 356 notes](#), also taught by Louigi!

I Discrete-Time Markov Chains

Conditional expectations will be important in this course. Recall $\mathbb{E}[X|Y = y_0]$, where X, Y are random variables. If Y is continuous, writing $\mathbb{E}[X|Y = y_0] = \frac{\mathbb{P}(X, Y=y_0)}{\mathbb{P}(Y=y_0)}$, will not work. Instead, we consider the slice of the joint density function $f(x, y)$ at $y = y_0$. The result is a one dimensional function $g(x)$ which may not have probability 1. Hence, we divide by $\int g(x)$ to make it into a density function:

$$\mathbb{E}[X|Y = y_0] = \int_{\mathbb{R}} \frac{f(x, y_0)}{\int_{\mathbb{R}} f(x, y_0) dx} x dx$$

DEF 1.1 We frequently write $f_{X|Y}(x) = f(x, y) / \int_{\mathbb{R}} f(x, y) dx$, and call this the *conditional density* of X given Y . For fixed y , then, $\mathbb{E}[X|Y = y] = \mathbb{E}[Z]$, where $Z \sim f_{X|Y}$.

INTRODUCTION

Before providing definitions, we give some examples of stochastic processes:

Eg. 1.1 A simple random walk: $S_{i+1} = S_i + X_i$, where $X_i \sim \text{Ber}(p)$ and $S_0 = 0$. We might ask: does S_i ever return to 0, i.e.

$$\mathbb{P}(\exists i > 0 : S_i = 0)$$

Eg. 1.2 A branching process: as in asexual reproduction, we have an initial node. Each node n has a number of children X_n , where $\frac{X_n}{2} \sim \text{Ber}(p)$. We denote Z_i to be the number of individuals in the i -th generation. We might ask: does Z_i ever have no children, i.e.

$$\mathbb{P}(\exists i > 0 : Z_i = 0)$$

Eg. 1.3 Choose k independent random points in the square $[0, \sqrt{k}]^2$. On average, then, there is 1 point within any unit square $U \subseteq [0, \sqrt{k}]^2$.

DEF 1.2 Given a finite or countable set V , a *Markov chain* with *state space* V is a sequence $X_n : n \geq 0$ of random variables, with $X_n \in V$, such that:

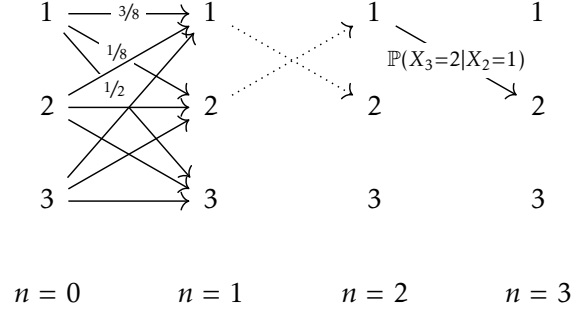
DEF 1.3

$$\underbrace{\mathbb{P}(X_{n+1} = v_{n+1})}_{\text{future}} \mid \underbrace{X_0 = v_0, \dots, X_{n-1} = v_{n-1}}_{\text{past}}, \underbrace{X_n = v_n}_{\text{present}} = \mathbb{P}(X_{n+1} = v_{n+1} \mid X_n = v_n)$$

In other words, the future only depends on the past via the present. This is called the

DEF 1.4 *Markov property*.

Sometimes we allow Markov chains to be only finitely large (i.e. $0 \leq n \leq m$). For instance, we limit ourselves to one weekend of gambling in Las Vegas. A graphical example would look something like:



We can decompose walks on Markov chains by discrete "steps":

PROP 1.1

$$\mathbb{P}(X_1 = v_1, \dots, X_n = v_n | X_0 = v_0) = \prod_{i=1}^n \mathbb{P}(X_i = v_i | X_{i-1} = v_{i-1})$$

PROOF.

$$\begin{aligned} & \mathbb{P}(X_1 = v_1, \dots, X_n = v_n | X_0 = v_0) \\ &= \mathbb{P}(X_2 = v_2, \dots, X_n = v_n | X_0 = v_0, X_1 = v_1) \mathbb{P}(X_1 = v_1 | X_0 = v_0) \quad \text{by cond. prob.} \\ & \vdots \\ &= \mathbb{P}(X_1 = v_1 | X_0 = v_0) \cdot \mathbb{P}(X_2 = v_2 | X_0 = v_0, X_1 = v_1) \cdots \mathbb{P}(X_n = v_n | X_0 = v_0, \dots, X_{n-1} = v_{n-1}) \\ &= \mathbb{P}(X_1 = v_1 | X_0 = v_0) \cdot \mathbb{P}(X_2 = v_2 | X_1 = v_1) \cdots \mathbb{P}(X_n = v_n | X_{n-1} = v_{n-1}) \quad \text{by Markov prop} \end{aligned}$$

□

Time-Homogeneous Markov Chains

We often write
THMC

We say that a Markov chain is *time-homogeneous* if, for all $u, v \in V$ and $n \geq 0$

DEF 1.5

$$\mathbb{P}(X_{n+1} = v | X_n = u) = \mathbb{P}(X_1 = v | X_0 = u)$$

In other words, the chain's behavior is described entirely by $\mathbb{P}(X_1 = v | X_0 = u)$ for each $(v, u) \in V \times V$. In this case, we can describe such probabilities in a *transition matrix* P :

DEF 1.6

$$P = (p_{u,v})_{(u,v) \in V^2} = (\mathbb{P}(X_1 = v | X_0 = u))_{(u,v) \in V^2}$$

Eg. 1.4 Recall the game Snakes and Ladders. A 6×6 grid is indexed $1, \dots, 36$. Players start at the 1 cell. The game ends when a player reaches the 36 cell. A die roll dictates how many spots one advances. There are some directed edges between cells (increasing: "ladders", decreasing: "snakes"). One must follow these

edges when one lands at its tail. Suppose a ladder exists from 11 to 27. Then

$$\mathbb{P}(X_{11} = 27 | X_{10} = 6, X_9 = 3) = \frac{1}{6} = \mathbb{P}(X_{11} = 27 | X_{10} = 6) = \mathbb{P}(X_2 = 27 | X_1 = 6)$$

We see that Snakes and Ladders is naturally modeled as a time-homogeneous Markov chain.

Eg. 1.5 Sampling without replacement is *not* a Markov chain. If we sample from $|X| = 10$, we have

$$\mathbb{P}(X_3 = a | X_2 = b) = 1/9$$

$$\mathbb{P}(X_3 = a | X_2 = b, X_1 = c) = 1/8$$

$$\mathbb{P}(X_3 = a | X_2 = b, X_1 = a) = 0$$

so we do not satisfy the Markov property.

Eg. 1.6 Returning to the Snakes and Ladders example, consider $S \subseteq V$. Let $T_S = \inf\{n \geq 0 : X_n \in S\}$, which we call the "*hitting time*" of S . We may ask...

- What is the average number of rounds to finite? We can write this as $\mathbb{E}[T_{\{36\}} | X_0 = 1]$.
- What is the probability of landing on 18 or 19 before the game ends? We can write this as $\mathbb{P}(T_{\{18,19\}} < T_{\{36\}} | X_0 = 1)$.
- What is the average number of visits to 6 before the game ends? We can write this as

$$\mathbb{E}[\#\{n \in [T_{\{36\}}] : X_n = 6\} | X_0 = 1]$$

- What is the expected proportion of time spent on state 5 before the game ends?
- If we allow two players, what is the probability that player 1 wins? Is this still a Markov chain?

DEF 1.7

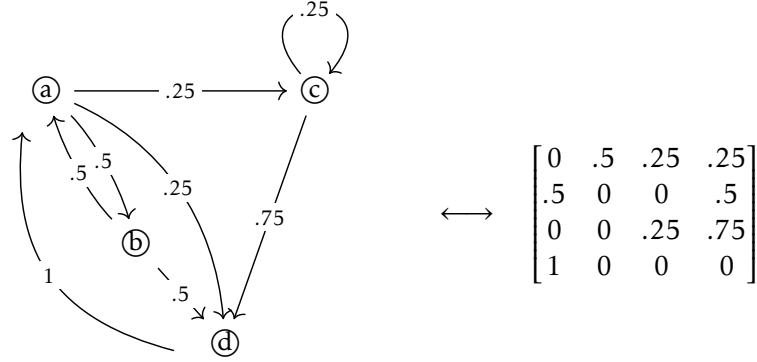
DEF 1.8 A matrix $P = (p_{u,v})_{(u,v) \in V^2}$ is called a *stochastic matrix* if every row sums to 1, i.e.

$$\forall u \in V, \sum_{v \in V} p_{u,v} = 1$$

Note that any stochastic matrix is the transition matrix of some time-homogeneous Markov chain with state space V and transition probabilities

$$\mathbb{P}(X_{n+1} = v | X_n = u) = \mathbb{P}(X_1 = v | X_0 = u) = p_{u,v}$$

A directed graph, together with its stochastic matrix, can visualize any THMC:



Eg. 1.7 Random walks on an undirected weighted graph, where edge weights dictate the proportional probability of transitioning between two states, are a special class of THMCs. In particular, given a graph $G = (V, E)$ with weights $w(e) > 0 : e \in E$, we set

$$p_{u,v} = \frac{w(\{u, v\})}{\sum_{z \in N(u)} w(\{u, z\})}$$

If there are no edges $u \leftrightarrow v$, we write $p_{u,v} = 0$.

Not every THMC can be represented by a random walk on an undirected weighted graph. In particular, see the directed graph listed above, or any transition matrix which is not symmetric.

As a concrete example, we can consider a random walk on the number line $\mathbb{Z}$, where, if $w(k, k+1) = \alpha$, $w(k-1, k) = \frac{\alpha}{2}$.

$$\dots \frac{1}{16} \quad -3 \quad \frac{1}{8} \quad -2 \quad \frac{1}{4} \quad -1 \quad \frac{1}{2} \quad 0 \quad \frac{1}{2} \quad 1 \quad 2 \quad \frac{4}{2} \quad 3 \quad \frac{8}{2} \quad \dots$$

Multi-Step Transition Probabilities

Given a THMC $X = X_n : n \geq 0$ with a transition matrix P , we write

$$\begin{aligned} \mathbb{P}(X_2 = v | X_0 = u) &= \sum_{w \in V} \mathbb{P}(X_2 = v, X_1 = w | X_0 = u) \\ &= \sum_{w \in V} \mathbb{P}(X_1 = w | X_0 = u) \mathbb{P}(X_2 = v | X_1 = w, \cancel{X_0 = u}) \quad \text{by Markov property} \\ &= \sum_{w \in V} p_{u,w} p_{w,v} = (P^2)_{u,v} \quad \text{or write } P_{u,v}^2 \end{aligned}$$

Hence, to determine a two-step transition probability, and by extension an n -step transition probability from u to w , we consider $P_{u,v}^n$. PROP 1.2

PROOF.

See [Prop 1.1](#) to expand probabilities, using Bayes', as needed. We get that

$$\begin{aligned}\mathbb{P}(X_n = v | X_0 = u) &= \sum_{v_1, \dots, v_{n-1} \in V} \mathbb{P}(X_1 = v_1, \dots, X_{n-1} = v_{n-1}, X_n = v | X_0 = u) \\ &= \sum_{v_1, \dots, v_{n-1} \in V} p_{u, v_1} p(v_1, v_2) \cdots p(v_{n-1}, v) = (P^n)_{u, v} \quad \square\end{aligned}$$

PROP 1.3 Thus, if P is a stochastic matrix, then so is P^n .

PROOF.

$$\sum_{v \in V} P^n_{u, v} = \sum_{v \in V} \mathbb{P}(X_n = v | X_0 = u) = 1. \quad \square$$

Theorem 1.1 Markov Property

If $X_n : n \geq 0$ is a THMC with state space V , then for all $u_0, \dots, u_{n-1}, u, v \in V$,

$$\mathbb{P}(X_{n+m} = v | X_0 = u_0, \dots, X_{n-1} = u_{n-1}, X_n = u) = \mathbb{P}(X_{n+m} = v | X_n = u) = P^m_{u, v}$$

PROOF.

One shows this by combining the Markov property with [Prop 1.2](#). $\square$

We *also* call this the Markov property.

DEF 1.9

We say that a Markov chain has an *initial distribution* $\alpha = (\alpha_v : v \in V)$ if $\mathbb{P}(X_0 = v) = \alpha_v$ for each $v \in V$. If this is the case, we often write α as a subscript of our state probabilities. For instance,

$$\mathbb{P}_\alpha(X_n = v) = \sum_{u \in V} \mathbb{P}_\alpha(X_0 = u, X_n = v) = \sum_{u \in V} \mathbb{P}_\alpha(X_0 = u) \mathbb{P}_\alpha(X_n = v | X_0 = u) = \sum_{u \in V} \alpha_u P^n_{u, v}$$

In particular, [Thm 1.1](#) is the Markov property combined with a claim about $P^n_{u, v}$

PROP 1.4 For an event E , depending only on $X_0, \dots, X_n$, with $\mathbb{P}(X_n = u, E) > 0$, we have

$$\mathbb{P}(X_{n+m} = v | X_n = u, E) = P^m_{u, v}$$

PROOF.

For any such event E , we can determine whether E occurs exactly when we know the realized values u_i of X_i for $i = 1, \dots, n-1$. Hence, we may write $\mathcal{S}$ to be the set of tuples $(u_0, \dots, u_{n-1})$ that guarantee E . It follows that

$$\mathbb{P}(X_n = u, E) = \sum_{\mathbf{s} \in \mathcal{S}} \mathbb{P}(\mathbf{X} = \mathbf{s}, X_n = u)$$

Similarly, we have

$$\begin{aligned}
 \mathbb{P}(X_{n+m} = v, X_n = u, E) &= \sum_{\mathbf{s} \in \mathcal{S}} \mathbb{P}(X_{n+m} = v, X_n = u, \mathbf{X} = \mathbf{s}) \\
 &= \sum_{\mathbf{s} \in \mathcal{S}} \mathbb{P}(X_{n+m} = v | X_n = u, \mathbf{X} = \mathbf{s}) \mathbb{P}(X_n = u, \mathbf{X} = \mathbf{s}) \\
 &= P_{u,v}^m \sum_{\mathbf{s} \in \mathcal{S}} \mathbb{P}(X_n = u, \mathbf{X} = \mathbf{s}) = P_{u,v}^m \mathbb{P}(X_n = u, E)
 \end{aligned}$$

Divide and use Bayes, and the result follows. $\square$

If X is a THMC with transition matrix P , then, for all $k \geq 1$, $X_{kn} : n \geq 0$ is a THMC with transition matrix P^k . PROP 1.5

For any $n \neq 0$, any sequence $u_0, \dots, u_{n+1} \in V$ satisfies PROOF.

$$\mathbb{P}(X_{(n+1)k} = u_{n+1} | X_0 = u_0, X_k = u_1, \dots, X_{nk} = u_n) = P_{u_n, u_{n+1}}^k \quad \square$$

Theorem 1.2 Chapman-Kolmogorov

For any Markov chain X with state space V , any $m, n \geq 0$, and $u, v \in V$,

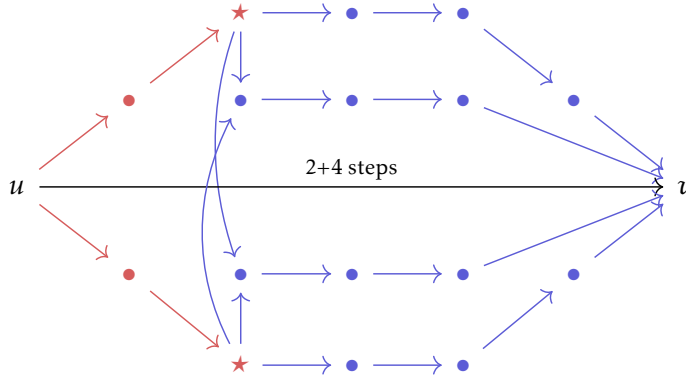
$$\mathbb{P}(X_{m+n} = v | X_0 = u) = \sum_{w \in V} \mathbb{P}(X_n = w | X_0 = u) \mathbb{P}(X_{m+n} = v | X_n = w)$$

If the X is time homogeneous, then this is $P_{u,v}^{n+m}$, which agrees with [Prop 1.2](#).

$$\begin{aligned}
 \mathbb{P}(X_{m+n} = v | X_0 = u) &= \sum_{w \in V} \mathbb{P}(X_{m+n} = v, X_n = w | X_0 = u) \\
 &= \sum_{w \in V} \mathbb{P}(X_{m+n} = v | X_n = w, X_0 = u) \mathbb{P}(X_n = w | X_0 = u) \\
 &= \sum_{w \in V} \mathbb{P}(X_{m+n} = v | X_n = w) \mathbb{P}(X_n = w | X_0 = u)
 \end{aligned}$$

Below is a visualization of the Chapman-Kolmogorov equation: to get from u to v over 6 steps, we pick some partition (e.g. $2 + 4 = 6$), and sum the probabilities of reaching v via

length 2 (red) and length 4 (blue) paths.



You should count 6 such combinations in the sample below.

STATIONARY AND LIMITING DISTRIBUTIONS

DEF 1.10 Recall from probability the *law of large numbers*: if $Y_n : n \geq 1$ are IID with common mean μ , then $\frac{S_n}{n} \rightarrow \mu$ in probability, where $S_n = \sum_{i=1}^n Y_i$, i.e. $\forall \varepsilon > 0$,

$$\limsup_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{S_n}{n} - \mu \right| > \varepsilon \right) = 0$$

If $Y_i \in \mathbb{Z}$ is a THMC, for $k, \ell \in \mathbb{Z}$ and $i \in [n-1]$,

$$\begin{aligned} P_{k,\ell} &:= \mathbb{P}(S_{n+1} = \ell | S_n = k, S_i = u_i \forall i) = \mathbb{P}(Y_{n+1} = \ell - k | S_n = k, S_i = u_i \forall i) \\ &= \mathbb{P}(Y_{n+1} = \ell - k | Y_1 = u_1 - u_0, Y_2 = u_2 - u_1, \dots, Y_n = k - u_{n-1}) \\ &= \mathbb{P}(Y_{n+1} = \ell - k) = \mathbb{P}(Y_1 = \ell - k) \end{aligned}$$

We conclude that $S_n : n \geq 0$ may be viewed as a THMC, with transition matrix P .

From now on, we denote by $\mathbb{P}_v(E)$ the probability $\mathbb{P}(E|v)$.

Eg. 1.8 A general two-state chain, with states A and B , can be described by

$$P = \begin{pmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{pmatrix}$$

Let $q_n = \mathbb{P}_A(X_n = A) = \mathbb{P}(X_n = A | X_0 = A)$. Then

$$\begin{aligned} q_{n+1} &= \mathbb{P}_A(X_{n+1} = A, X_n = A) + \mathbb{P}_A(X_{n+1} = A, X_n = B) \\ &= \mathbb{P}_A(X_{n+1} = A | X_n = A) \mathbb{P}_A(X_n = A) + \mathbb{P}_A(X_{n+1} = A | X_n = B) \mathbb{P}_A(X_n = B) \\ &= (1 - \alpha)q_n + \beta(1 - q_n) = \beta + (1 - \alpha - \beta)q_n \end{aligned}$$

This recurrence has a unique solution. In particular, one can find

$$q_n = \frac{\beta}{\alpha + \beta} + (1 - \alpha - \beta)^n \frac{\alpha}{\alpha + \beta}$$

It follows that $q_n \rightarrow \frac{\beta}{\alpha+\beta}$, and hence $\mathbb{P}_A(X_n = B) = 1 - q_n \rightarrow \frac{\alpha}{\alpha+\beta}$. Likewise:

$$\mathbb{P}_B(X_n = B) = \frac{\alpha}{\alpha+\beta} + (1 - \beta)^n \frac{\beta}{\alpha+\beta}$$

So $\mathbb{P}_B(X_n = B) \rightarrow \frac{\alpha}{\alpha+\beta}$.

Let $\pi := (\pi_A, \pi_B) := \left(\frac{\beta}{\alpha+\beta}, \frac{\alpha}{\alpha+\beta}\right)$ be the distribution of our initial state X_0 . Then

$$\mathbb{P}_\pi(X_1 = A) = \pi_A \mathbb{P}_A(X_1 = A) + \pi_B \mathbb{P}_B(X_1 = A) = \pi_A$$

and, similarly, $\mathbb{P}_\pi(X_1 = B) = \pi_B$. Hence, if X_0 has initial distribution π , then X_1 also has distribution π . By induction, X_n has distribution $\pi \forall n \geq 0$.

When we say $X = \text{Markov}(P)$, we mean that X is a THMC with transition matrix P .

A probability distribution π is called **stationary** if $\pi P = \pi$. Similarly, a probability distribution λ is called a **limiting distribution** if, for each $u, v \in V$

DEF 1.11

DEF 1.12

$$(P^n)_{u,v} \rightarrow \lambda_v \text{ as } n \rightarrow \infty$$

In other words, $\mathbb{P}_u(X_n = v) \rightarrow \lambda_v$. Note that, for any initial distribution α , we have $\alpha P^n \rightarrow \lambda$, i.e. $(\alpha P^n)_v \rightarrow \lambda_v$, where λ is limiting. As proof:

PROOF.

$$(\alpha P^n)_v = \sum_{u \in V} \alpha_u P^n_{u,v} \rightarrow \sum_{u \in V} \alpha_u \lambda_v = \lambda_v$$

□

If λ is a limiting distribution for P , then λ is stationary for P .

PROP 1.6

Fix any initial distribution α , we have

PROOF.

$$\lambda = \lim_{n \rightarrow \infty} (\alpha P^n) = \lim_{n \rightarrow \infty} (\alpha P^{n-1} P) = \left(\lim_{n \rightarrow \infty} \alpha P^{n-1} \right) P = \lambda P \quad \square$$

Stationary distributions need not be unique, but limiting distributions are (as the limit $\lim_{n \rightarrow \infty} \alpha P^n$ is well-defined). In general, then, stationary distributions need not be limiting distributions.

A stochastic matrix P is called **regular** if $\exists n \geq 1$ such that $P^n > 0$ on all entries.

DEF 1.13

Theorem 1.3 Fundamental Theorem of Markov Chains

Every finite, regular stochastic matrix P has a limiting distribution π .

Incorporating some of the formulations above, this is equivalent to saying: *For a regular stochastic matrix, there exists a unique distribution $\pi = (\pi_v : v \in V)$ such that $\pi P = \pi$ and $\mathbb{P}_u(X_n = v) \rightarrow \pi_v \forall u, v \in V$.*

When $n = 0$, $P^n = I$, which encapsulates the idea that, at timestep 0, we will be at our initial positions.

In this case, there is a unique stationary distribution, and it is the unique limiting distribution.

A stationary distribution always exists!

Let $\rho = \langle 1, \dots, 1 \rangle$. Then note that $P\rho = \rho$, since the sum of any row in P must be 1. Hence, P has eigenvalue 1. It follows that it has a left eigenvector, i.e. $\pi : \pi P = \pi$. This is exactly a stationary distribution, as long as we scale suitably such that π is a distribution.

However, then process of scaling into a distribution is non-trivial. Since π may have negative coordinates, and hence $\sum \pi_i = 0$, we must consider instead $|\pi|$, i.e. prove it is also an eigenvalue.

This is true, but requires the fact that P is stochastic

Periodicity of States

DEF 1.14 For $u, v \in V$, we say that v is *accessible* from u if $\exists n \geq 0$ such that $(P^n)_{u,v} > 0$. Equivalently, in the directed graph generated by P , there is a directed path from u to v . When v is accessible from u , we write $u \rightarrow v$.

DEF 1.15 States u and v *communicate* if $u \rightarrow v$ and $v \rightarrow u$. When u and v communicate, we write $u \leftrightarrow v$. Observe that communication is an equivalence relation. Hence, the state space V can be written as a disjoint union of mutually-communicating states, called a

DEF 1.16 *communication class*. Note that, in the directed graph generated by P , these correspond to the strongly connected components.

DEF 1.17 We say that P is *irreducible* if there is only one communication class.

Clearly, if P is regular, then it is irreducible

PROP 1.7 $u \rightarrow v \iff \mathbb{P}_u(T_{\{v\}} < \infty) > 0$.

DEF 1.18 The *period* of a state $u \in V$ is

$$d(u) := \gcd(n > 0 : P_{u,u}^n > 0)$$

DEF 1.19 If $d(u) = 1$, we call u *aperiodic*. By extension, P is aperiodic if $d(u) = 1 \forall u \in V$, and X is aperiodic if $X = \text{Markov}(P)$ for P aperiodic.

PROP 1.8 If $u \leftrightarrow v$, then $d(u) = d(v)$.

PROOF.

Let $I = \{n > 0 : P_{u,u}^n > 0\}$, and similarly J for v . Hence, $d(u) = \gcd(I)$ and $d(v) = \gcd(J)$. Let $a, b > 0$ such that $P_{u,v}^a > 0$ and $P_{v,u}^b > 0$. Then

$$P_{u,u}^{a+b} \geq P_{u,v}^a P_{v,u}^b > 0$$

$\implies a + b \in I$, so $d(u) | a + b$. Now, if $n \in J$, then

$$P_{u,u}^{a+b+n} \geq P_{u,v}^a P_{v,v}^n P_{v,u}^b > 0$$

$\implies a + b + n \in I$, so $d(u) | n + a + b$. But, by the previous line, $d(u) | n$. Since $n \in J$ is arbitrary, we can write $d(u) | \gcd(J) = d(v)$.

Symmetrically, we could conclude that $d(v) | d(u)$, so indeed $d(v) = d(u)$. $\square$

PROP 1.9 Let $I = \{n > 0 : P_{u,u}^n > 0\}$. If $\gcd(I) = 1$, then $\exists a, b \in I$ such that $\gcd(a, b) = 1$.

This is not true for any I (and thus relies not only on number theory). Let $\ell, m \in I$, with $\ell < m$. Let $k = m - \ell$. If $k = 1$, then $\gcd(\ell, m) = 1$. Otherwise, since $\gcd(I) = 1$, there is an $n \in I$ with $k \nmid n$. We then write $n = qk + r$, with $r \in [k - 1]$. Then $m' \in (q + 1)m \in I$, since $P_{u,u}^{(q+1)m} \geq (P_{u,u}^m)^{q+1}$. Symmetrically, we can argue $\ell' = (q + 1)\ell \in I$.

Similarly, $\ell^* := \ell' + n \in I$, since $P_{u,u}^{\ell'+n} \geq P_{u,u}^{\ell'} P_{u,u}^n$. We have

$$\begin{aligned} m' - \ell^* &= (q + 1)m - (q + 1)\ell - n = (q + 1)(m - \ell) - n \\ &= (q + 1)k - n = k - r \in [k - 1] \end{aligned}$$

TODO...

PROOF.

Theorem 1.4 Postage Stamp Lemma

If P is irreducible and aperiodic, then $\forall u, v \in V, \exists N$ such that $P_{u,v}^n > 0 \forall n \geq N$.

Before proving this, we note that, for $a, b \geq 1$ with $\gcd(a, b) = 1$, then for any $q \geq ab$, we can write $q = ja + kb$ for integers $j, k \geq 0$.

Fix $u, v \in V$. Since P is aperiodic, there are $a, b \geq 1$ with $P_{u,u}^a, P_{u,u}^b > 0$ and $\gcd(a, b) = 1$, by [Prop 1.9](#). Since P is irreducible, there is some $m > 0$ with $P_{u,v}^m > 0$. Thus, let $N = m + ab$. For any $n \geq N$, let $q = n - m$. We have that $q \geq ab$, so we can find $j, k \geq 0$ with $q = ja + kb$. Then

$$P_{u,v}^n = P_{u,v}^{q+m} = P_{u,v}^{ja+kb+m} \geq P_{u,u}^{ja} P_{u,u}^{kb} P_{u,v}^m \geq (P_{u,u}^a)^j (P_{u,u}^b)^k P_{u,v}^m$$

All are positive, so $P_{u,v}^n > 0$, as desired.

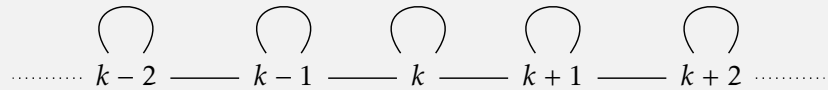
PROOF.

Theorem 1.5 Characterization of Regular Markov Chains

Let $P = (p_{u,v})_{u,v \in V}$ be a stochastic matrix, where $|V| < \infty$. Then

$$P \text{ is regular} \iff P \text{ is irreducible and aperiodic}$$

We first note why finiteness is necessary. Consider:



with all edges having weight 1. This graph is clearly aperiodic and irreducible, but not regular.

($\implies$) If P is regular, then $\exists n > 0$ s.t. $P_{u,v}^n > 0$ for all $u, v \in V$. Then, for all $u, v \in V$, we have $u \rightarrow v$ and $v \rightarrow u$. Hence, P is irreducible. Now, if P is irreducible, then for all $u \in V$, there is some $v \in V$ such that $P_{v,u} > 0$ (think about this in graph theoretic

PROOF.

terms). Then, let $n > 0$ be such that $P_{u,u}^n$ is positive. We have

$$P_{u,u}^{n+1} \geq P_{u,v}^n P_{u,v} > 0$$

So, with $I = \{m > 0 : P_{u,u}^m > 0\}$, $d(u) = \gcd(I) \leq \gcd(n, n+1) = 1$. It follows that $d(u) = 1$, so u is aperiodic (and hence P is aperiodic).

($\Leftarrow$) By [Thm 1.4](#), for each $u, v \in V$, there exists $N : P_{u,v}^n > 0 \forall n \geq N$. Let N^* be the maximum value of N determined over all pairs $(u, v) \in V^2$. Then, for $n \geq N^*$ and all $u, v \in V$, $P_{u,v}^n > 0$. It follows that all entries of P^n are positive, and we are done. $\square$

Finding Stationary Distributions

Recall that $x = (x_v : v \in V)$ is a stationary distribution if $xP = x$. Let V be finite. Then, for a stationary distribution x , we have

$$\begin{aligned} x_1 p_{1,1} + \cdots + x_n p_{n,1} &= x_1 \\ x_1 p_{1,2} + \cdots + x_n p_{n,2} &= x_2 \\ &\vdots \\ x_1 p_{1,n} + \cdots + x_n p_{n,n} &= x_n \end{aligned}$$

We have n equations, n unknowns, and a homogeneous system, so there is not a unique solution. If

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \end{pmatrix}$$

We can compute $x = \langle t, 2t, 2t \rangle$. But, noting that x is a probability distribution, and hence $5t = 1$, this yields $x = \langle 1/5, 2/5, 2/5 \rangle$. We'll consider some special cases.

UNDIRECTED GRAPHS

Let $G = (V, E)$ be undirected. Then we define a THMC by

$$P_{u,v} = \begin{cases} \frac{1}{\deg(u)} & \{u, v\} \in E \\ 0 & \text{o.w.} \end{cases}$$

This is distinct from [Example 1.7](#)

Let $x = (\deg(v) : v \in V)$. We have

$$\begin{aligned} (xP)_v &= \sum_{u \in V} \deg(u) P_{u,v} = \sum_{u \in N(v)} \deg(u) \cdot \frac{1}{\deg(u)} \\ &= \deg(v) \end{aligned}$$

Hence, $xP = x$. Recalling that $\sum_{v \in V} \deg(v) = 2|E|$, we conclude that

$$\left(\frac{\deg(v)}{2|E|} : v \in V \right)$$

is a stationary distribution.

UNDIRECTED WEIGHTED GRAPHS

This is not distinct from [Example 1.7](#)

Let $G = (V, E)$ be undirected. Then, we define a THMC by

$$P_{u,v} = \begin{cases} \frac{w(\{u,v\})}{\sum_{z \in N(u)} w(\{u,z\})} & v \in N(u) \\ 0 & \text{o.w.} \end{cases}$$

Let $x = (\sum_{e: e \ni v} w(e) : v \in V)$. Then we can compute $xP = x$, and similar to above,

$$x = \left(\frac{\sum_{e: e \ni v} w(e)}{2 \sum_{e \in E} w(e)} : v \in V \right)$$

is a stationary distribution.

Transience and Recurrence

Recall $T_S = \inf\{n \geq 0 : X_n \in S\}$, the "hitting time" of S . We let $R_S = \inf\{n > 0 : X_n \in S\}$. Note that if $T_S > 0$, $T_S = R_S$. Otherwise, R_S gives the first "*return time*" to the set S .

DEF 1.20

A state $v \in V$ is called *recurrent* if $\mathbb{P}_v(R_{\{v\}} < \infty) = 1$. If all states of v are recurrent, we may call P and $X = \text{Markov}(P)$ recurrent. Otherwise, we call v *transient*, and similarly extend the notion to the transition matrix and chain when all state are transient.

DEF 1.21

DEF 1.22

For a given state $v \in V$, we call $L_v = |\{n \geq 0 : X_n = v\}|$ the *local time* of v . Note that, if $v = X_j$ and v is recurrent, then $L_v = \infty$.

DEF 1.23

Intuitively, if $L_v > k$ when $X_0 = v$, then $L_v > k - 1$ when $X_{i_1} = v$, where i_1 is the first time we return to v .

Let $X = \text{Markov}(P)$. For any state $v \in V$ and $k > 1$,

PROP 1.10

$$\mathbb{P}_v(L_v > k) = \mathbb{P}_v(L_v > 1)^k$$

Using the law of total probability:

PROOF.

$$\begin{aligned} \mathbb{P}_v(L_v > k) &= \mathbb{E}[\mathbb{P}_v(L_v > k | R_v)] = \sum_{t=1}^{\infty} \mathbb{P}_v(R_v = t) \mathbb{P}_v(L_v > k | R_v = t) \\ &= \sum_{t=1}^{\infty} \mathbb{P}_v(R_v = t) \mathbb{P}_v(L_v > k | R_v = t, X_t = v) \\ &= \sum_{t=1}^{\infty} \mathbb{P}_v(R_v = t) \mathbb{P}_v(L_v > k - 1) \\ &= \mathbb{P}_v(L_v > k - 1) \sum_{t=1}^{\infty} \mathbb{P}_v(R_v = t) \\ &= \mathbb{P}_v(L_v > k - 1) \mathbb{P}_v(R_v < \infty) = \mathbb{P}_v(L_v > k - 1) \mathbb{P}_v(L_v > 1) \end{aligned}$$

The result follows by induction. □

PROP 1.11

$$\mathbb{P}_v(L_v = \infty) = \begin{cases} 1 & v \text{ recurrent} \\ 0 & v \text{ transient} \end{cases}$$

PROOF.

This follows directly from [Prop 1.10](#) + monotonicity of probability. $\square$

PROP 1.12

$$\sum_{n=0}^{\infty} P_{v,v}^n = \begin{cases} \infty & v \text{ recurrent} \\ \frac{1}{1-\mathbb{P}_v(L_v > 1)} & v \text{ transient} \end{cases} = \begin{cases} \infty & v \text{ recurrent} \\ \frac{1}{1-\mathbb{P}_v(R_{\{v\}} < \infty)} & v \text{ transient} \end{cases}$$

PROOF.

This follows from linearity of expectation, and the fact that, for a non-negative integer variable X , $\mathbb{E}[X] = \sum_{i=0}^{\infty} \mathbb{P}(X > i)$ (We use, and prove, this fact later).

$$\begin{aligned} \sum_{n=0}^{\infty} P_{v,v}^n &= \sum_{n=0}^{\infty} \mathbb{P}_v(X_n = v) = \sum_{i=0}^{\infty} \mathbb{E}[\mathbb{1}_{X_i=v}] \\ &= \mathbb{E}\left[\sum_{i=0}^{\infty} \mathbb{1}_{X_i=v}\right] = \mathbb{E}_v[L_v] = \sum_{i=0}^{\infty} \mathbb{P}(L_v > i) \\ &= \sum_{i=0}^{\infty} \mathbb{P}_v(L_v > 1)^i = \begin{cases} \infty & X \text{ recurrent} \\ \frac{1}{1-\mathbb{P}_v(L_v > 1)} & X \text{ transient} \end{cases} \end{aligned}$$

Note that $\mathbb{P}_v(L_v > 1) \equiv \mathbb{P}_v(R_v < \infty)$ $\square$

PROP 1.13 If $u \leftrightarrow v$, then u is transient $\iff v$ is transient.

PROOF.

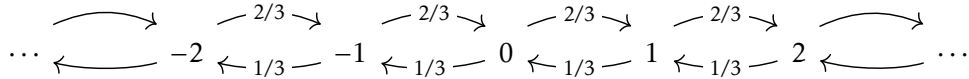
Fix $a, b \geq 0$ with $P_{u,v}^a, P_{v,u}^b > 0$. Then

$$\begin{aligned} \sum_{n=0}^{\infty} P_{v,v}^n &\geq \sum_{n=0}^{\infty} P_{v,v}^{a+b+n} = \sum_{n=0}^{\infty} P_{v,u}^b P_{u,u}^n P_{u,v}^a \\ &= P_{v,u}^b P_{u,v}^a \sum_{n=0}^{\infty} P_{u,u}^n \end{aligned}$$

Thus, if v is transient, then $\sum_{n=0}^{\infty} P_{v,v}^n < \infty$, so it must be that $\sum_{n=0}^{\infty} P_{u,u}^n < \infty$, i.e. u is transient. The argument is identical in reverse. $\square$

Eg. 1.9 If $u \leftrightarrow v$ and u is recurrent, then $\mathbb{P}_u(T_{\{v\}} < \infty) = 1$.

Eg. 1.10 The following chain is completely transient:



In fact, we could replace $2/3$ by p , and $1/3$ by $1-p$. In this case, the chain is irreducible. To see that it is transient, we have

$$P_{0,0}^{2n} = \binom{2n}{n} p^n (1-p)^n$$

Then

$$\sum_{n=0}^{\infty} P_{0,0}^{2n} < \sum_{n=0}^{\infty} 2^{2n} p^n (1-p)^n = \sum_{n=0}^{\infty} (4p(1-p))^n < \infty \quad \text{if } p \neq \frac{1}{2}$$

By [Prop 1.12](#). Notice that $\sum_{n=0}^{\infty} P_{0,0}^n = \sum_{n=0}^{\infty} P_{0,0}^{2n}$, since it is only possible to return on even-length cycles.

We conclude that the chain is transient when $p \neq \frac{1}{2}$.

Stirling's Formula provides

$$m! \sim \left(\frac{m}{e}\right)^m \sqrt{2\pi m}$$

FACT

in that

$$\lim_{m \rightarrow \infty} \frac{m!}{\left(\frac{m}{e}\right)^m \sqrt{2\pi m}} = 1$$

These facts, though out of the scope of this course, can be derived from a careful treatment of Reimann sums

This fact implies

$$e \left(\frac{n}{e}\right)^n \leq n! \leq \frac{e(n+1)}{4} \left(\frac{n+1}{e}\right)^n$$

Fig. 1.11 We return to the previous example, letting $p = \frac{1}{2}$. Then

$$\begin{aligned} \sum_{n=0}^{\infty} P_{0,0}^{2n} &= \sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2} [p(1-p)]^n \sim \sum_{n=0}^{\infty} \frac{\left(\frac{2n}{e}\right)^{2n} \sqrt{2\pi(2n)}}{\left[\left(\frac{n}{e}\right)^n \sqrt{2\pi n}\right]^2} [p(1-p)]^n \\ &= \sum_{n=0}^{\infty} \frac{[4p(1-p)]^n}{\sqrt{\pi n}} = \sum_{n=0}^{\infty} \frac{1}{\sqrt{\pi n}} = \infty \end{aligned}$$

We conclude that the chain is recurrent when $p = \frac{1}{2}$.

If V is finite, then there is at least one recurrent state.

PROP 1.14

Fix an initial distribution $\alpha = (\alpha_v : v \in V)$. Then $\mathbb{P}_\alpha(\sum_{v \in V} L_v = \infty) = 1$. We conclude

PROOF.

that there is at least one state $v \in V$ with $\mathbb{P}_\alpha(L_v = \infty) > 0$. But also:

$$\begin{aligned} \mathbb{P}_\alpha(L_v = \infty) &= \sum_{n=0}^{\infty} \mathbb{P}_\alpha(L_v = \infty, T_v = n) = \sum_{n=0}^{\infty} \mathbb{P}_\alpha(L_v = \infty | T_v = n) \mathbb{P}_\alpha(T_v = n) \\ &= \sum_{n=0}^{\infty} \mathbb{P}_\alpha(L_v = \infty | T_v = n, X_n = v) \mathbb{P}_\alpha(T_v = n) \\ &= \sum_{n=0}^{\infty} \mathbb{P}_\alpha(L_v = \infty | X_n = v) \mathbb{P}_\alpha(T_v = n) \\ &= \sum_{n=0}^{\infty} \mathbb{P}_v(L_v = \infty) \mathbb{P}_\alpha(T_v = n) \end{aligned}$$

So $\mathbb{P}_v(L_v = \infty) > 0 \implies \mathbb{P}_v(L_v = \infty) = 1$, by [Prop 1.11](#). $\square$

PROP 1.15 Finite, irreducible chains are recurrent.

PROOF.

Since the chain is finite it has at least one recurrent state, by [Prop 1.14](#). Then all states must be recurrent, since the chain is irreducible, by [Prop 1.13](#). $\square$

Canonical Decompositions

Fix a transition matrix P and list the communication classes of V as

$$D_1, D_2, \dots \quad (\text{transient}) \quad C_1, C_2, \dots \quad (\text{recurrent})$$

DEF 1.24 Note that we can split the chain up in this way by [Prop 1.13](#). Set $D = \cup_{i \geq 0} D_i$. Then the *canonical decomposition* of the chain is defined to be

$$D \sqcup C_1 \sqcup C_2 \sqcup \dots$$

DEF 1.25 We say that a communication class C is *closed* if, for any $u \in C, v \notin C, p_{u,v} = 0$. Intuitively, if $X_0 \in C$, or we enter C at some later time, we will never leave C .

PROP 1.16 If C is a recurrent communication class, then C is closed.

PROOF.

Fix $u \in C, v \notin C$. Suppose $v \mapsto u$. If $p_{u,v} > 0$, then $u \mapsto v$, so $v \in C$ $\nmid$. Suppose $v \not\mapsto u$. Then $\mathbb{P}_u(R_u = \infty) \geq \mathbb{P}_u(X_1 = v) = p_{u,v}$. But $\mathbb{P}_u(R_u = \infty) = 0$, since u is recurrent. It follows that $p_{u,v} = 0$. $\square$

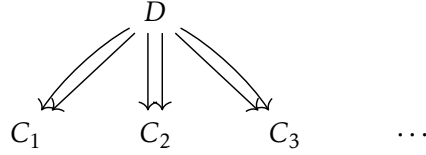
The converse of [Prop 1.16](#) is not true in generality, but it is in the finite case:

PROP 1.17 Finite, closed communication classes are recurrent.

PROOF.

From any starting state in C , we must visit some state $u \in C$ infinitely often, as $|C| < \infty$ and $X_t \in C \forall t$. But recurrence is a class property by [Prop 1.13](#). Hence, all of C is recurrent. $\square$

When our communication classes are closed, we have



Proof of Fundamental Theorem of Markov Chains

Recall [Thm 1.3](#):

Every finite, regular stochastic matrix P has a limiting distribution π .

We will prove this in two steps. First, we will find some stationary distribution. Then, we will prove that this is a limiting distribution.

Theorem 1.6 Existence Theorem

Let P be irreducible and finite. Let $(X_n : n \geq 0) = \text{Markov}(P)$. Fix $u \in V$, which we call a reference vertex, and, for any $v \in V$, define

$$\gamma_v = \mathbb{E}_u[|\{0 \leq n < R_u : X_n = v\}|]$$

Let $\gamma = (\gamma_v : v \in V)$. Then $\gamma P = \gamma$, and $0 < \gamma_v < \infty \forall v \in V$.

Alternatively,
irreducible and
recurrent

By $\mathbb{E}_u$, we mean the
expectation, under
the assumption that
 $X_0 = u$

Observe that $\gamma_u = 1$. Write

$$\gamma_v = \mathbb{E}_u \left[\sum_{n=0}^{R_u-1} \mathbb{1}_{X_n=v} \right] = \mathbb{E}_u \left[\sum_{n=1}^{R_u} \mathbb{1}_{X_n=v} \right] = \mathbb{E}_u \left[\sum_{n=1}^{\infty} \mathbb{1}_{X_n=v} \mathbb{1}_{n \leq R_u} \right]$$

We utilize a second indicator variable in order to use linearity of expectation (otherwise, our sum would index over a random variable, which is not valid). Then

$$\gamma_v = \sum_{n=1}^{\infty} \mathbb{E}_u[\mathbb{1}_{X_n=v} \mathbb{1}_{n \leq R_u}] = \sum_{n=1}^{\infty} \mathbb{P}_u(X_n = v, n \leq R_u)$$

Now, using the law of total probability,

$$\begin{aligned} \mathbb{P}_u(X_n = v, n \leq R_u) &= \sum_{w \in V} \mathbb{P}_u(X_{n-1} = w, X_n = v, n \leq R_u) \\ &= \sum_{w \in W} \mathbb{P}_u(X_n = v | X_{n-1} = w, n \leq R_u) \mathbb{P}_u(X_{n-1} = w, n \leq R_u) \\ &= \sum_{w \in V} \mathbb{P}_u(X_{n-1} = w, n \leq R_u) P_{w,v} \end{aligned}$$

PROOF.

So

$$\begin{aligned}
 \gamma_v &= \sum_{n=1}^{\infty} \sum_{w \in W} P_{w,v} \mathbb{P}_u(X_{n-1} = w, n \leq R_u) = \sum_{w \in V} P_{w,v} \left(\sum_{n=1}^{\infty} \mathbb{P}_u(X_{n-1} = w, n-1 < R_u) \right) \\
 &= \sum_{w \in V} P_{w,v} \sum_{n=0}^{\infty} \mathbb{P}_u(X_n = w, n < R_u) = \sum_{w \in V} P_{w,v} \sum_{n=0}^{\infty} \mathbb{E}_u[\mathbb{1}_{X_n=w} \mathbb{1}_{n < R_u}] \\
 &= \sum_{w \in V} P_{w,v} \mathbb{E}_u \left[\sum_{n=0}^{R_u-1} \mathbb{1}_{X_n=w} \right] = \sum_{w \in V} P_{w,v} \gamma_w = (\gamma P)_v
 \end{aligned}$$

$\Rightarrow \gamma P = \gamma$. Furthermore, $\gamma_v > 0$, since $u \mapsto v$. Letting $n : P_{v,u}^n > 0$, we have $\gamma_u = (\gamma P^n)_u \geq \gamma_v P_{v,u}^n$. Noting that $\gamma_u = 1$, this shows $\gamma_v < \infty$. $\square$

PROP 1.18 If P is irreducible and $\mathbb{E}_u[R_u]$ is finite, then $\pi = (\pi_v : v \in V)$, where

$$\pi_v = \frac{\gamma_v}{\sum_{w \in V} \gamma_w}$$

is a stationary distribution.

PROOF.

Observe

$$\sum_{w \in V} \gamma_w = \sum_{w \in V} \mathbb{E}_i \left[\sum_{n=0}^{R_i-1} \mathbb{1}_{X_n=w} \right] = \mathbb{E}_i \left[\sum_{n=0}^{R_i-1} \sum_{w \in V} \mathbb{1}_{X_n=w} \right] = \mathbb{E}_i[R_i]$$

So we require that $\mathbb{E}_u[R_u] < \infty$, then a stationary distribution exists. $\square$

PROP 1.19 If P is irreducible and $\mathbb{E}_u[R_u]$ is finite, and π is stationary for P , then $\pi_u = \frac{1}{\mathbb{E}_u[R_u]}$.

An exercise, if u is recurrent and $u \mapsto v$, then $\mathbb{P}_u(T_v < \infty) = 1 \ \forall v \in V$.

PROP 1.20 π , as defined above, is a limiting distribution.

PROOF.

Consider two independent copies $X_n, Y_n = \text{Markov}(P)$. Then (X_n, Y_n) is a Markov chain with transition matrix $Q = (q_{(v,w),(x,y)})_{(v,w),(x,y) \in V \times V}$. In particular,

$$q_{(v,w),(x,y)} = p_{v,x} p_{w,y}$$

Fix some state $u \in V$. Let α be the initial distribution that has $X_0 = u$ and $Y_0 \sim \pi$.

$$\alpha_{(x,y)} = \begin{cases} \pi(y) & x = u \\ 0 & \text{o.w.} \end{cases}$$

Remark that, as π is stationary, $Y_n \sim \pi \ \forall n \geq 0$. For any $u \in V$, we'd like that $P_{u,v}^n \rightarrow \pi(v)$.

Let $M = \inf\{n \geq 0 : X_n = Y_n\}$ be the first meeting time of the X_n and Y_n chains. If P is

finite and regular, then Q is finite and regular. Then

$$\mathbb{P}_\alpha(M < \infty) \geq \mathbb{P}_\alpha(T_{(v,v)} < \infty) = \sum_{(x,y) \in V \times V} \alpha(x,y) \mathbb{P}_{(x,y)}(T_{v,v} < \infty) = 1$$

It follows that $\lim_{n \rightarrow \infty} \mathbb{P}_\alpha(M > n) = 0$. We claim that $\mathbb{P}_\alpha(X_n = v, M \leq n) = \mathbb{P}_\alpha(Y_n = v, M \leq n) \forall n \geq 0$. Assuming this, then

$$\begin{aligned} \mathbb{P}_u(X_n = v) &= \mathbb{P}_\alpha(X_n = v) = \mathbb{P}_\alpha(X_n = v, M \leq n) + \mathbb{P}_\alpha(X_n = v, M > n) \\ \mathbb{P}_\pi(Y_n = v) &= \mathbb{P}_\alpha(Y_n = v) = \mathbb{P}_\alpha(Y_n = v, M \leq n) + \mathbb{P}_\alpha(Y_n = v, M > n) \\ \implies \mathbb{P}_u(X_n = v) - \pi(v) &= \mathbb{P}_\alpha(X_n = v, M > n) - \mathbb{P}_\alpha(Y_n = v, M > n) \\ \implies |\mathbb{P}_u(X_n = v) - \pi(v)| &= |\mathbb{P}_\alpha(X_n = v, M > n) - \mathbb{P}_\alpha(Y_n = v, M > n)| \leq \mathbb{P}_\alpha(M > n) \end{aligned}$$

But $\mathbb{P}_\alpha(M > n) \rightarrow 0$, so it must be that $\mathbb{P}_u(X_n = v) \rightarrow \pi(v)$.

We still must show the claim, however.

$$\begin{aligned} \mathbb{P}_\alpha(X_n = v, M \leq n) &= \sum_{k=0}^n \sum_{w \in V} \mathbb{P}_\alpha(X_n = v, M = k, X_k = w) \\ &= \sum_{k=0}^n \sum_{w \in V} \mathbb{P}_\alpha(M = k, X_k = w) \mathbb{P}_\alpha(X_n = v | X_k = w, M = k) \\ &\stackrel{\text{MP}}{=} \sum_{k=0}^n \sum_{w \in V} \mathbb{P}_\alpha(M = k, X_k = w) P_{w,v}^{n-k} \\ &\text{and now we reverse!} \\ &= \sum_{k=0}^n \sum_{w \in V} \mathbb{P}_\alpha(M = k, Y_k = w) P_{w,v}^{n-k} \\ &\stackrel{\text{MP}}{=} \sum_{k=0}^n \sum_{w \in V} \mathbb{P}_\alpha(M = k, Y_k = w) \mathbb{P}_\alpha(X_n = v | Y_k = w, M = k) \\ &= \mathbb{P}_\alpha(Y_n = v, M \leq n) \quad \square \end{aligned}$$

HITTING TIMES AND ABSORBING STATES

We call a state $v \in V$ *absorbing* if $\mathbb{P}_v(X_1 = v) = 1$.

DEF 1.26

Eg. 1.12 Consider a game where we continually bet \$1 with winning probability p , until we earn \$ k or run out of money. Modeling the state space as our balance, we have absorbing states at \$0 and \$ k .

Observe that each absorbing state forms a recurrent communication class of size 1. As a consequence, we have a variant of the canonical form, where $A \subseteq V$ are absorbing states,

and we write $V = A \sqcup (V \setminus A)$. For a stochastic matrix P and block matrices Q, R , we have:

$$P = \begin{array}{c|c} V \setminus A & A \\ \hline Q & R \\ 0 & I_{|A|} \end{array} \Rightarrow P^2 = \begin{array}{c|c} V \setminus A & A \\ \hline Q^2 & QR + RI \\ 0 & I_{|A|} \end{array}$$

$$\Rightarrow P^n = \begin{array}{c|c} V \setminus A & A \\ \hline Q^n & (I_{|V \setminus A|} + Q + \cdots + Q^{n-1})R \\ 0 & I_{|A|} \end{array}$$

Schematically, then

$$\begin{array}{ccc} \begin{pmatrix} Q \end{pmatrix} & & \begin{pmatrix} I_{|A|} \end{pmatrix} \\ \downarrow & & \downarrow \\ V \setminus A & \xrightarrow{R} & A \end{array}$$

PROP 1.21 Let $c \in V$ be an absorbing state, where V is finite. Let Q be as above. Then, for $u \in V$,

$$\mathbb{E}_u(T_c) = \begin{cases} \infty & \exists b \text{ absorbing} : u \rightarrow b \\ \sum_{v \neq c \in V} (1 - Q)_{u,v}^{-1} & c \text{ is the only absorbing state, all other transient} \end{cases}$$

with $\mathbb{E}_\alpha(T_c) = \sum_{u \in V} \alpha_u \mathbb{E}_u(T_c)$.

PROOF.

Fix an initial distribution $\alpha = (\alpha_v : v \in V)$ and an absorbing state $c \in V$. Then

$$\mathbb{E}_\alpha(T_c) = \sum_{v \in V} \mathbb{P}_\alpha(X_0 = v) \mathbb{E}_\alpha(T_c | X_0 = v) = \sum_{v \in V} \alpha_v \mathbb{E}_v(T_c)$$

Recall from probability that $\mathbb{E}X = \sum_{n \geq 1} \mathbb{P}(X \geq n)$ whenever X is supported on the non-negative integers. Assume v is not absorbing. Then

$$\begin{aligned} \mathbb{E}_v(T_c) &= \sum_{n \geq 1} \mathbb{P}_v(T_c \geq n) = \sum_{n \geq 1} \sum_{w \neq c \in V} \mathbb{P}_v(X_{n-1} = w) = \sum_{n \geq 1} \sum_{w \neq c \in V} P_{v,w}^{n-1} \\ &= \sum_{n \geq 1} \sum_{w \in V \setminus A} Q_{w,v}^{n-1} + \sum_{n \geq 1} \sum_{w \neq c \in A} P_{v,w}^{n-1} \end{aligned}$$

where Q is as in the absorbing canonical form above. If there is an absorbing state $b \neq c$ such that $v \mapsto b$, then $\mathbb{E}_v(T_c) = \infty$. Otherwise, the second sum above is 0, and we get

$$\mathbb{E}_v(T_c) = \sum_{w \in V \setminus A} \sum_{n \geq 1} Q_{w,v}^{n-1} = \sum_{w \in V \setminus A} \sum_{n \geq 0} Q_{w,v}^n$$

Here, $I + Q + Q^2 + \cdots = (I - Q)^{-1}$. □

Why is this invertible?
Generally, if $A^n \rightarrow 0$ as $n \rightarrow \infty$, then $I - A$ is invertible.

Theorem 1.7 Hitting States in (Transient, Absorbing) Chains

Let P be a finite transition matrix, such that all states are either transient or absorbing. Let A be the absorbing states, with $|A| = m$. Then

$$P^n = \begin{bmatrix} Q^n & (I + Q + \cdots + Q^{n-1})R \\ 0 & I \end{bmatrix} \Rightarrow \lim_{n \rightarrow \infty} P^n = \begin{bmatrix} 0 & (I - Q)^{-1}R \\ 0 & I \end{bmatrix}$$

Furthermore, for any transient states u, v ,

$$\mathbb{E}_u(L_v) = (I - Q)_{u,v}^{-1} \quad \text{and} \quad \mathbb{E}_u[T_A] = \sum_{v \in V \setminus A} (I - Q)_{u,v}^{-1}$$

Finally, for any transient state u and absorbing state c ,

$$\mathbb{P}_u(X_{T_A} = c) = ((I - Q)^{-1}R)_{u,c}$$

We'll first justify $\lim_{n \rightarrow \infty} Q^n = 0$. Suppose $u, v \in V \setminus A$ satisfy $\lim_{n \rightarrow \infty} Q_{u,v}^n > 0$, with $Q_{u,v}^n = \mathbb{P}_u(X_n = v)$. Then, since all states outside of $V \setminus A$ are absorbing, this implies that

$$\lim_{n \rightarrow \infty} \mathbb{P}_u(X_n \in V \setminus A) > 0 \implies \mathbb{P}_u(L_{V \setminus A} = \infty) > 0$$

But, since V is finite, there must be some $w \in V \setminus A$ with $\mathbb{P}_u(L_w = \infty) > 0$. Then, the Markov property holds that $\mathbb{P}_u(L_w = \infty) > 0$, a contradiction of transience.

For $\mathbb{E}_u(L_v)$, we have

$$\mathbb{E}_u(L_v) = \mathbb{E}_u \left(\sum_{n \geq 0} \mathbb{1}_{X_n=v} \right) = \sum_{n \geq 0} \mathbb{P}_u(X_n = v) = \sum_{n \geq 0} (Q^n)_{u,v} = (I - Q)_{u,v}^{-1}$$

as desired. For $\mathbb{E}_u[T_A]$, we argue as follows: the time T_A is equal to the number of times we visit $V \setminus A$ in all time, including $X_0 = u$, since we cannot return to $V \setminus A$ from A , as A is absorbing. Hence,

$$\mathbb{E}[T_A] = \sum_{v \in V \setminus A} \mathbb{E}_u(L_v) = \sum_{v \in V \setminus A} (I - Q)_{u,v}^{-1}$$

by the arguments above.

Finally, we consider $\mathbb{P}_u(X_{T_A} = c)$. This is equivariant to the probability, summing over all $n \geq 0$, that we hit a state $v \in V \setminus A$ in n steps, and proceed to hit c in the

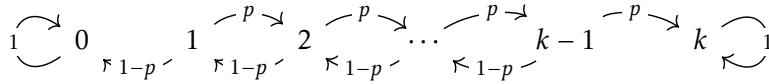
PROOF.

subsequent step. In other words,

$$\begin{aligned}\mathbb{P}_u(X_{T_A} = c) &= \sum_{n \geq 0} \sum_{v \in V \setminus A} \mathbb{P}_u(X_n = v) \mathbb{P}(X_{n+1} = c | X_n = v) = \sum_{n \geq 0} \sum_{v \in V \setminus A} (Q^n)_{u,v} R_{v,c} \\ &= \sum_{n \geq 0} (Q^n)_{(u)} \cdot R^{(c)} = \sum_{n \geq 0} (Q^n R)_{u,c} = \left(\left(\sum_{n \geq 0} Q^n \right) R \right)_{u,c} = ((1 - Q)^{-1} R)_{u,c}\end{aligned}$$

as desired. $\square$

Fig. 1.13 Consider Gambler's Ruin, with reward limit k :



Then $A = \{0, k\}$. Let $p_i = \mathbb{P}_i(X_{T_A} = k)$. Computing manually, this is $p_i = p p_{i+1} + (1-p)p_{i-1}$, with boundary conditions $p_0 = 0$ and $p_k = 1$. We conclude that

$$p_i = \begin{cases} \frac{1 - \left(\frac{1-p}{p}\right)^i}{1 - \left(\frac{1-p}{p}\right)^k} & p \neq \frac{1}{2} \\ \frac{i}{k} & p = \frac{1}{2} \end{cases}$$

Similarly, we manually compute $\mathbb{E}_i(T_A) = 1 + p \mathbb{E}_{i+1}(T_A) + (1-p) \mathbb{E}_{i-1}(T_A)$, with boundary conditions $\mathbb{E}_0(T_A) = \mathbb{E}_k(T_A) = 0$. This gives

$$\mathbb{E}_i(T_A) = \begin{cases} \frac{k(1-r^i) - i(1-r^k)}{p(1-r)(1-r^k)} & p \neq \frac{1}{2} \\ i(k-i) & p = \frac{1}{2} \end{cases} \quad \text{where } r = \frac{1-p}{p}$$

REVERSIBILITY

Some motivation: on an atomic scale, we cannot pin-point an "arrow of time," meaning, it is impossible to distinguish between a video on the atomic scale and that video in reverse. Entropy is hard to gauge! For instance, consider the following chains:



By symmetry, it must be that $(\pi_1, \pi_2, \pi_3) = (1/3, 1/3, 1/3)$ is the stationary and limiting distribution for each chain. Can we tell if a "video" of the left chain is played in reverse? By symmetry, no. What about for the right video? We should observe "net counter-clockwise" behavior, as the clockwise transition from a given state only has probability $1/3$. Thus, we could distinguish between a video and its reverse.

Sometimes, it may only be possible to observe the arrow of time in a chain while close to initial conditions. The Ehrenfest Urn is a box with a partition down the middle, with particles initially on the left side of the box.

At each time step, we choose a random point in the box, and move it to its opposing side. When the urn reaches equilibrium, it will be impossible to distinguish an arrow of time. However, at the start, we expect left particles to move to the right.

These notions motivate the following:

Let P be a stochastic matrix with stationary distribution π . Then the *time reversal* of P is the matrix $\hat{P}$ with DEF 1.27

$$\pi_u \hat{P}_{u,v} = \pi_v P_{v,u} \quad \forall u, v \in V$$

In other words, when taking one step from the initial distribution, the probability of reaching state v from u in the time reversal is equal to the probability of reaching state u from v in the original chain. Since π is stationary, this means that the probability of seeing $u \mapsto v$ at any step in the chain is equal to that of seeing $v \mapsto u$ at this step.

A chain $X = \text{Markov}(P)$ and its stochastic matrix P are called *reversible* if $\hat{P} = P$.

DEF 1.28

Theorem 1.8 Norris' Theorem

Let P be irreducible with stationary distribution π . Let $X_0 \sim \pi$ and consider the chain $X_n : n \in [N] = \text{Markov}(P)$. Then $(Y_0, \dots, Y_N) := (X_N, \dots, X_0)$ is $\text{Markov}(\hat{P})$.

Not provided in class or examinable. See textbook.

□

PROOF.

Eg. 1.14 Consider the Ehrenfest Urn example. Let X_n denote the number of particles on the left container at time n . Suppose that there are k total particles, such that $V = [k]$. Then

$$\mathbb{P}(X_{n+1} = i + 1 | X_n = i) = \frac{k - i}{k} \quad \mathbb{P}(X_{n+1} = i - 1 | X_n = i) = \frac{i - 1}{k}$$

What might a stationary distribution be? Typically, a good strategy is to guess the limiting distribution (in this case, we have regularity, so this is sound). Assuming that each particle is equally likely to be on the left or right, we conclude that $X_n \sim \text{Bin}(k, \frac{1}{2})$, so $\pi(i) = \frac{1}{2^k} \binom{k}{i}$. Checking that this is stationary:

$$\begin{aligned} \mathbb{P}_\pi(X_1 = i) &= \pi(i - 1) \mathbb{P}_{i-1}(X_1 = i) + \pi(i + 1) \mathbb{P}_{i+1}(X_1 = i) \\ &= \frac{1}{2^k} \left[\binom{k}{i-1} \frac{k - (i-1)}{k} + \binom{k}{i+1} \frac{i+1}{k} \right] \\ &= \frac{1}{2^k} \left[\frac{(k-1)!}{i(k-i)!} \cdot i + \frac{(k-1)!}{i!(k-1)!} \cdot (k-1) \right] = \frac{1}{2^k} \binom{k}{i} = \pi_i \end{aligned}$$

We can now compute the time reversal matrix, based on [Def 1.27](#):

$$\hat{P}_{i,i+1} = \frac{\pi_{i+1}}{\pi_i} P_{i+1,i} = \frac{\binom{k}{i+1} i + 1}{\binom{k}{i}} = \frac{k-i}{k} = P_{i,i+1}$$

One shows a similar result for $P_{i,i-1}$, so $P = \hat{P}$, and P is reversible.

Theorem 1.9 Reversible Chains are Random Weighted Walks

Let P be a stochastic matrix, with a stationary distribution π . Then P is reversible if and only if P can be represented as a random walk on an undirected weighted graph, i.e. there exists a graph $G = (V, E)$ with weights $w : E \rightarrow [0, \infty)$ such that

$$P_{u,v} = \frac{w(\{u, v\})}{\sum_{w \in N(u)} w(\{u, w\})} \quad \forall u, v \in V$$

PROOF.

($\implies$) Let P be reversible with stationary distribution π . Let $G = (V, E)$ have edge weights $w(\{x, y\}) = \pi_x P_{x,y} = \pi_y P_{y,x}$, by reversibility. Then, for $u, v \in V$,

$$\frac{w(\{u, v\})}{\sum_{w \in N(u)} w(\{u, w\})} = \frac{\pi_u P_{u,v}}{\sum_{w \in N(u)} \pi_u P_{u,w}} = P_{u,v}$$

($\impliedby$) Fix $G = (V, E)$ with $w : E \rightarrow [0, \infty)$. Suppose P is given by a random walk on G :

$$P_{u,v} = \frac{w(\{u, v\})}{\sum_{w \in N(u)} w(\{u, w\})}$$

Then, for $x \in V$, a stationary distribution is provided by $\pi(x) = \frac{\sum_{e \in \delta(x)} w(e)}{2 \sum_{e \in E} w(e)}$. Notate $D := 2 \sum_{e \in E} w(e)$. For each $u, v \in V$,

$$\begin{aligned} \pi_u P_{u,v} &= \frac{\sum_{e \in \delta(u)} w(e)}{D} \cdot \frac{w(\{u, v\})}{\sum_{w \in N(u)} w(\{u, w\})} = \frac{w(\{u, v\})}{D} \\ &= \frac{\sum_{e \in \delta(v)} w(e)}{D} \cdot \frac{w(\{v, u\})}{\sum_{e \in \delta(v)} w(e)} = \pi_v P_{v,u} \end{aligned}$$

Hence, P is reversible. □

PROP 1.22 If π satisfies $\pi_u P_{u,v} = \pi_v P_{v,u} \quad \forall u, v \in V$, then π is stationary for P .

PROOF.

Check that

$$\mathbb{P}_\pi(X_1 = v) = \sum_{u \in V} \pi_u P_{u,v} = \sum_{u \in V} \pi_v P_{v,u} = \pi_v \implies \pi P = \pi \quad \square$$

Metropolis-Hastings

Consider a cipher of the alphabet, i.e. a bijection $f : C \rightarrow A$, where $A = \{a, b, \dots, z\}$, and C are a set of cipher symbols. From a collection of ciphered text, how can we determine f ?

One naive solution is to model the frequency of symbols in C , and "align" it with the frequency of the alphabet in English text. However, the mapping from letters to their frequency is not injective, so we cannot identify letters uniquely from their frequency.

Alternatively, we can let $M : A \times A \rightarrow [0, 1]$ describe the frequency of transitions, i.e. let

$$M(x, y) = \text{proportion of "x" that are followed by a "y" in English text}$$

For instance, $M(s, t)$ may be large, but $M(x, j) = 0$. Then for a given text $s = (s_1, \dots, s_L)$ of cipher characters, and a hypothesized mapping f ,

$$\text{Plausibility}(f) = \prod_{i=1}^{L-1} M(f(s_i), f(s_{i+1}))$$

This is exactly the probability that a Markov chain with transition matrix M produces the sequence $(f(s_1), \dots, f(s_L))$. We can maximize the plausibility score algorithmically:

```

function MONTE-CARLO
   $f \leftarrow$  any guess  $f : C \rightarrow A$ 
  while not satisfied do
    compute  $\text{Pl}(f)$ 
    generate  $f_*$  from  $f$  by a random transposition
    if  $\text{Pl}(f_*) > \text{Pl}(f)$  then
       $f \leftarrow f_*$ 
    else
      toss coin
      if coin =  $H$  then
        Do nothing
      else
         $f \leftarrow f_*$ 

```

This algorithm, the *Monte-Carlo algorithm*, is a close cousin of Metropolis-Hastings, which we'll cover now.

DEF 1.29

Let V be a finite state space. Let $\pi = (\pi_v)_{v \in V}$ be a probability distribution on states, which we do not have direct access to. Let T be a transition matrix such that $T_{x,y} > 0 \iff T_{y,x} > 0$. We wish to use T to generate a sample from π . To do so, we can build a new transition matrix P with $\pi P = \pi$, in which case it suffices to show that $\pi_u P_{u,v} = \pi_v P_{v,u}$, by [Prop 1.22](#). At this point, we can sample* from π by taking long walks on P .

*"sample" in a loose sense. One can take a running average of steps, or, if P is normal, take the final destination

For $x, y \in V$, let

$$A(x, y) := \frac{\pi_y T_{y,x}}{\pi_x T_{x,y}}$$

Note that $A(x, y) = \frac{1}{A(y, x)}$. Then, we define

$$P_{x,y} := \begin{cases} T_{x,y} & x \neq y, A(x, y) \geq 1 \\ T_{x,y}A(x, y) & x \neq y, A(x, y) < 1 \\ T_{x,y} + \sum_{z:A(x,z)<1} T_{x,z}(1 - A(x, z)) & x = y \end{cases}$$

Intuitively, this asks "how much more likely is it that I encounter $y \mapsto x$ versus $x \mapsto y$ on my first transition."

PROP 1.23 P , as defined, is a stochastic matrix.

PROOF.

$$\begin{aligned} \sum_{y \in V} P_{x,y} &= \left[\sum_{y \neq x: A(x,y) \geq 1} T_{x,y} \right] + \left[\sum_{y \neq x: A(x,y) < 1} T_{x,y}A(x, y) + T_{x,x} + \sum_{z:A(x,z)<1} T_{x,z}(1 - A(x, z)) \right] \\ &= \sum_{y \in V} T_{x,y} = 1 \end{aligned}$$

□

Algorithmically, P behaves the following way:

DEF 1.30 **function** *METROPOLIS-HASTINGS*($X_n = x$)
 choose proposal state y with probability $\mathbb{P}_x(X_1 = y) = T_{x,y}$.
 $U \leftarrow \text{Unif}[0, 1]$
 if $U \leq A(x, y)$ **then**
 | $X_{n+1} \leftarrow y$
 else
 | $X_{n+1} \leftarrow x$

In other words, the probability of accepting the proposal is $A(x, y)$.

Hey... so "A" for accepting

PROP 1.24 P is reversible with stationary distribution π .

PROOF.

Fix $x, y \in V$. We show that $\pi_x P_{x,y} = \pi_y P_{y,x}$. If $x = y$, this is clear. Otherwise, assume $A(x, y) \geq 1$. Then

$$\pi_x P_{x,y} = \pi_x T_{x,y} = \frac{\pi_y T_{y,x}}{A(x, y)} = \pi_y T_{y,x} A(y, x) = \pi_y P_{y,x}$$

□

The algorithm, or, equivalently, the definition for P , is called the Metropolis-Hastings algorithm, or "rejection sampling."

A similar subject to look into is "mixing times"

II Branching Processes

This topic covers a special case of Markov chains, called branching processes.

Eg. 2.1 Let $X \sim \text{Ber}(\frac{1}{2})$. Consider a root node. If $X = H$, we give that node 2 children. Else, we give it no children. We repeat this process for each child.

This is a model for asexual reproduction (or, alternatively, one's surname passing down patrilineally).

We can generalize this slightly by having the number of children of parent i in generation n be distributed according to $B_{n,i} \sim B$, where $B \in \mathbb{Z}_{\geq 0}$ are IID. We let Z_n denote the number of total children at generation n .

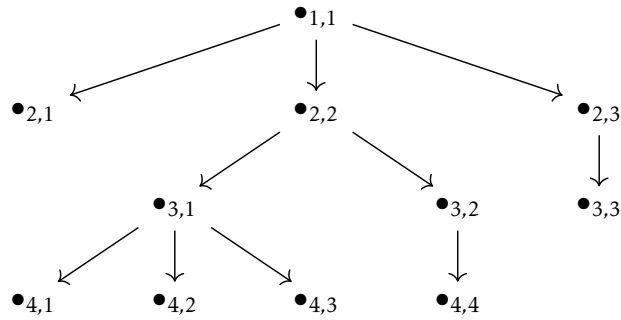
We can represent a realized sampling from B as a matrix of non-negative integers, where $B_{n,i} = (B)_{n,i}$. If generation $n-1$ has Z_{n-1} individuals, then the next generation has

$$B_{n,1} + B_{n,2} + \dots + B_{n,Z_{n-1}}$$

individuals. For instance, consider

$$B = \begin{bmatrix} 3 & 1 & 0 & 0 & \dots \\ 0 & 2 & 1 & 4 & \dots \\ 3 & 1 & 0 & 1 & \dots \\ 0 & 0 & 0 & 0 & \dots \\ 2 & 1 & 2 & 3 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Then, if we consider only the first initial child,



Let $B_{n,i} \sim B$ be IID, non-negative, integer random variables. Let Z_0 be some fixed integer. Inductively define $Z_n = \sum_{i=1}^{Z_{n-1}} B_{n,i}$. Then Z_n is called a *branching process*, and we write $Z_n \sim \text{Bienaymé}(B)$.

DEF 2.1

Branching processes are time-homogeneous Markov chain, with transition matrix

PROP 2.1

$$P_{k,\ell} = \mathbb{P}(B_{1,1} + \dots + B_{1,k} = \ell)$$

PROOF.

Our state space consists of non-negative integers. We verify the Markov property:

$$\begin{aligned}\mathbb{P}(Z_{n-1} = \ell | Z_0 = s_0, \dots, Z_{n-1} = s_{n-1}, Z_n = k) &= \mathbb{P}(B_{n+1,1} + \dots + B_{n+1,k} = \ell | \dots) \\ &= \mathbb{P}(B_{n+1,1} + \dots + B_{n+1,k} = \ell) \\ &= \mathbb{P}(B_{1,1} + \dots + B_{1,l} = \ell) =: \clubsuit\end{aligned}$$

Then, we have

$$\begin{aligned}\mathbb{P}(Z_{n+1} = \ell | Z_n = k) &= \sum_{s \geq 0} \mathbb{P}(Z_0 = s_0, \dots, Z_{n-1} = s_{n-1}, Z_{n+1} = \ell | Z_n = k) \\ &= \sum_{s \geq 0} \mathbb{P}(Z_0 = s_0, \dots, Z_{n-1} = s_{n-1} | Z_n = k) \\ &\quad \cdot \mathbb{P}(Z_{n+1} = \ell | Z_n = k, Z_0 = s_0, \dots, Z_{n-1} = s_{n-1}) \\ &= \sum_{s \geq 0} \mathbb{P}(Z_0 = s_0, \dots, Z_{n-1} = s_{n-1} | Z_n = k) \clubsuit = \clubsuit\end{aligned}$$

So, indeed, Z_n is time-homogeneous. By $\clubsuit$, we have the transition matrix

$$P_{k,\ell} = \mathbb{P}(B_{1,1} + \dots + B_{1,k} = \ell) \quad \square$$

DEF 2.2 We are curious as to the survival or extinction of Z_n . We say that Z_n *survives* if $Z_n > 0 \forall n \geq$

DEF 2.3 0, or in probability, $\mathbb{P}(Z_n > 0 \forall n \geq 0) = 1$. We call it *extinct* otherwise.

PROP 2.2 If $Z_n \sim \text{Bienaymé}(B)$, then

$$\mathbb{E}[Z_n | Z_0 = k] = k(\mathbb{E}[B])^n$$

PROOF.

We proceed by induction on n . When $n = 0$, then $\mathbb{E}[Z_0 | Z_0 = k] = k = k(\mathbb{E}[B])^0$.

Now, we have

$$\begin{aligned}\mathbb{E}[Z_{n+1} | Z_n = k] &= \mathbb{E}[B_{n+1,1} + \dots + B_{n+1,k} | Z_n = k] \\ &= \sum_{i=1}^k \mathbb{E}[B_{n+1,i} | Z_n = k] = k\mathbb{E}[B]\end{aligned}$$

Thus, $\mathbb{E}[Z_{n+1} | Z_n] = Z_n \mathbb{E}[B]$. Then, using the tower law of expectation,

$$\begin{aligned}\mathbb{E}[Z_{n+1}] &= \mathbb{E}_{Z_n}[\mathbb{E}[Z_{n+1} | Z_n]] = \mathbb{E}_{Z_n}[Z_n \mathbb{E}[B]] \\ &= \mathbb{E}[Z_n] \mathbb{E}[B] = k(\mathbb{E}[B])^n \mathbb{E}[B] = k(\mathbb{E}[B])^{n+1}\end{aligned}$$

using induction. □

PROP 2.3 If $\mathbb{E}[B] < 1$, then $\mathbb{P}(\text{survival} | Z_0 = k) = 0 \forall k \geq 0$.

Fix $\varepsilon > 0$. Let $N \geq 0$ be large enough such that $k(\mathbb{E}[B])^N < \varepsilon$. Then

PROOF.

$$\begin{aligned} \mathbb{P}(\text{survival} | Z_0 = k) &\leq \mathbb{P}(Z_N \geq 1 | Z_0 = k) \\ &\leq \sum_{\ell \geq 1} \mathbb{P}(Z_N \geq \ell | Z_0 = k) = \mathbb{E}[Z_N | Z_0 = k] = k(\mathbb{E}[B])^N < \varepsilon \quad \square \end{aligned}$$

Generating Functions

Let X be a random variable. Recall from probability the *moment generating function* G_X , DEF 2.4 where

$$G_X(s) = \mathbb{E}[s^X]$$

For an integer-valued random variable,

$$G_X(s) = \sum_{k \in \mathbb{Z}} s^k \mathbb{P}(X = k)$$

and, similarly, for a continuous random variable,

$$G_X(s) = \int_{\mathbb{R}} s^x f_X(x) dx$$

We will focus on integer-valued variables for branching processes.

Eg. 2.2 When X is integer-valued,

$$G_X(1) = \mathbb{E}[1^X] = \sum_{k \in \mathbb{Z}} 1^k \mathbb{P}(X = k) = 1$$

Similarly,

$$|G_X(s)| = |\mathbb{E}[s^X]| = \left| \sum_{k \in \mathbb{Z}} s^k \mathbb{P}(X = k) \right| \leq \sum_{j \in \mathbb{Z}} |s|^j \mathbb{P}(X = j) \leq G_X(1)$$

We conclude that $G_X(s)$ is defined for all $|s| \leq 1$. (In fact, it is infinitely differentiable on this interval).

Eg. 2.3 Let $X = \text{Ber}(p)$. Then $\mathbb{E}[s^X] = (1 - p) \cdot 1 + ps = 1 - p + ps$. For kX , $\mathbb{E}[s^{kX}]$ becomes $1 - p + ps^k$

Eg. 2.4 Let $X = \text{Bin}(n, p) = B_1 + \dots + B_n : B_i \sim \text{Ber}(p)$. Then

$$\mathbb{E}[s^X] = \mathbb{E}[s^{B_1 + \dots + B_n}] = \prod_{i=1}^n \mathbb{E}[s^{B_i}] = [1 - p + ps]^n$$

Considering the special case $\text{Bin}(n, \frac{\lambda}{n})$, where $\lambda > 0$, and we think of n as large,

$$\mathbb{E}[s^X] = \left(1 - \frac{\lambda}{n} + \frac{\lambda s}{n}\right)^n = \left(1 + \frac{\lambda(s-1)}{n}\right)^n \approx e^{\lambda(s-1)}$$

In some textbooks, this is $\mathbb{E}[e^{sX}]$, but these notions are equivalent up to a change of variables

Eg. 2.5 When $X = \text{Poisson}(\lambda)$, where $\mathbb{P}(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$,

$$\mathbb{E}[s^X] = \sum_{k \geq 0} \frac{s^k e^{-\lambda} \lambda^k}{k!} = e^{-\lambda} \sum_{k \geq 0} \frac{(\lambda s)^k}{k!} = e^{-\lambda} e^{\lambda s} = e^{\lambda(s-1)}$$

We conclude that the moments of a Poisson variable can be estimated by a suitable binomial variable.

Eg. 2.6 Suppose $X_n = A_{n,1} + \dots + A_{n,n}$, where $A_{n,i}$ are independent Bernoulli (not necessarily identically distributed), such that $\mathbb{E}[X_n] = \lambda$. We can ask: under what conditions on the expectation of $\mathbb{E}[A_{n,i}]$ does $G_{X_n}(s) \rightarrow e^{\lambda(s-1)}$?

We now connect these ideas to branching processes. Recall that an offspring distribution B is a non-negative integer-valued random variable.

Eg. 2.7 Let $X \sim 2\text{Ber}(p)$. When $p = 1/3$, we have that $G_X(s) = \frac{2}{3} + \frac{1}{3}s^2$. Check these graphs for $p = 1/2$ and $p = 1/3$.

We want to get an idea of how G_B looks on $s \in [0, 1]$, where B is an offspring distribution:

$$\begin{aligned} G_B(0) &= 0^0 \mathbb{P}(B=0) + \sum_{k \geq 1} 0^k \mathbb{P}(B=k) &&= \mathbb{P}(B=0) \\ G_B(1) &&&= 1 \\ G_B(s) &= \sum_{k \geq 0} s^k \mathbb{P}(B=k) &&\geq 0 \text{ for } s \geq 0 \\ G'_B(s) &= \sum_{k \geq 0} k s^{k-1} \mathbb{P}(B=k) = \sum_{k \geq 1} k s^{k-1} \mathbb{P}(B=k) &&\geq 0 \text{ for } s \geq 0 \\ G'_B(0) &= 1 \cdot 0^0 \mathbb{P}(B=1) + \sum_{k \geq 2} k 0^{k-1} \mathbb{P}(B=k) &&= \mathbb{P}(B=1) \\ G'_B(1) &= \sum_{k \geq 1} k \cdot 1^{k-1} \mathbb{P}(B=k) = \sum_{k \geq 0} k \mathbb{P}(B=k) &&= \mathbb{E}[B] \\ G''_B(s) &= \sum_{k \geq 0} k(k-1) s^{k-2} \mathbb{P}(B=k) = \sum_{k \geq 2} k(k-1) s^{k-2} \mathbb{P}(B=k) &&\geq 0, \text{ and } > 0 \text{ when } \mathbb{P}(B \geq 2) > 0 \end{aligned}$$

Alternatively, $G''_B(s) > 0$ when $\mathbb{P}(B \leq 1) < 1$.

Thus, we can characterize $G_B(s)$ as a positive, non-decreasing, convex function, starting at $\mathbb{P}(B=0)$ and ending at 1 for $s \in [0, 1]$. In particular, there exists some $0 \leq s < 1$ with $G_B(s) = s$ if and only if $\mathbb{E}[B] > 1$ or $\mathbb{P}(B=1) = 1$

PROP 2.4 If $Z_n : n \geq 0$ is Bienaymé(B) with $Z_0 = 1$, then, for $n \geq 1$, we have

$$G_{Z_n}(s) = \underbrace{G_B(G_B(\dots(G_B(s))))}_{n \text{ times}}$$

We'll show this by induction. When $n = 1$, $G_{Z_1} = G_{B_{1,1}}(s) = G_B(s)$.

PROOF.

Now, for $n \geq 1$, we have

$$G_{Z_{n+1}} = \mathbb{E}[s^{Z_{n+1}}] = \sum_{k \geq 0} \mathbb{P}(Z_n = k) \mathbb{E}[s^{Z_{n+1}} | Z_n = k] \text{ by tower property}$$

Then, we have

$$\begin{aligned} \mathbb{E}[s^{Z_{n+1}} | Z_n = k] &= \mathbb{E}[s^{B_{n+1,1} + \dots + B_{n+1,k}} | Z_n = k] = \mathbb{E}[s^{B_{n+1,1} + \dots + B_{n+1,k}}] \\ &= \prod_{i=1}^k \mathbb{E}[s^{B_{n+1,i}}] \text{ by independence} \\ &= G_B(s)^k \end{aligned}$$

Then, continuing from above,

$$\sum_{k \geq 0} \mathbb{P}(Z_n = k) \mathbb{E}[s^{Z_{n+1}} | Z_n = k] = \sum_{k \geq 0} \mathbb{P}(Z_n = k) G_B(s)^k = \mathbb{E}[G_B(s)^{Z_n}] = G_{Z_n}(G_B(s))$$

and, applying induction, we're done. $\square$

As a corollary,

$$\mathbb{P}(Z_n = 0 | Z_0 = 1) = G_B(G_B(\dots(G_B(0))))$$

PROP 2.5

We know $\mathbb{P}_1(Z_n = 0) = G_{Z_n}(0)$. $\square$

PROOF.

Write $\tau = \inf(n \geq 0 : Z_n = 0)$, i.e. the extinction time of Z_n . Note that $\tau = \infty$ if and only if $Z_n > 0 \forall n \geq 0$. Then $\mathbb{P}_1(\tau < \infty) = \lim_{n \rightarrow \infty} \mathbb{P}_1(Z_n = 0)$.

PROP 2.6

This is an application of the monotone limit theorem. First we take a detour in probability. For events $(E_n : n \geq 1)$ such that $E_1 \subseteq E_2 \subseteq \dots$, we have

PROOF.

$$\begin{aligned} \mathbb{P}(\cup_{n \geq 1} E_n) &= \mathbb{P}(E_1 \cup (E_2 \setminus E_1) \cup \dots \cup (E_n \setminus E_{n-1})) \\ &= \mathbb{P}(E_1) + \sum_{n \geq 2} \mathbb{P}(E_n \setminus E_{n-1}) = \mathbb{P}(E_1) + \lim_{N \rightarrow \infty} \sum_{n=2}^N \mathbb{P}(E_n \setminus E_{n-1}) \\ &= \lim_{N \rightarrow \infty} \mathbb{P}(E_N) \quad \text{by additivity} \end{aligned}$$

Setting $E_n = \{Z_n = 0\}$ for $n \geq 1$ (note that these events are nested), we conclude

$$\mathbb{P}(\cup_{n \geq 1} \{Z_n = 0\}) = \lim_{n \rightarrow \infty} \mathbb{P}(Z_n = 0)$$

But $\cup_{n \geq 1} \{Z_n = 0\}$ is the event $\tau < \infty$, and we are done. $\square$

Theorem 2.1 Fundamental Theorem of Branching Processes

Let $(Z_n : n \geq 0)$ be Bienaymé(B), with $Z_0 = 1$. Then

$$\mathbb{P}_1(\text{extinction}) = \mathbb{P}_1(\tau < \infty)$$

is the smallest non-negative solution s of $G_B(s) = s$. In particular, $\mathbb{P}_1(\tau < \infty) < 1$ if and only if either $\mathbb{E}[B] > 1$ or $\mathbb{P}(B = 1) = 1$.

PROOF.

Let $s := \mathbb{P}_1(\tau < \infty)$. We've shown that $G_{Z_n}(0) = \mathbb{P}_1(Z_n = 0) \rightarrow s$ as $n \rightarrow \infty$, by [Prop 2.6](#). But, since G_B is continuous on $[0, 1]$, we also have that $G_B(G_{Z_n}(0)) \rightarrow G(s)$ as $n \rightarrow \infty$. But $G_B(G_{Z_n}(0)) = \mathbb{P}_1(Z_{n+1} = 0)$ by [Prop 2.5](#), so it must also converge to s . Thus, $G_B(s) = s$.

It remains to show that s is the smallest solution to $G(s) = s$. In particular, suppose $t \geq 0$ satisfies $G_B(t) = t$. Then $t \geq 0$, so $G'_B \geq 0$, and thus, by monotonicity, $G_B(t) \geq G_B(0)$. But then $t \geq G_B(0)$. Again by monotonicity, $G_B(t) \geq G_B(G_B(0))$, i.e. $t \geq G_B(G_B(0))$. By induction, and [Prop 2.5](#), $t \geq \lim_{n \rightarrow \infty} \mathbb{P}_1(Z_n = 0) = \mathbb{P}_1(\tau < \infty) = s$. $\square$

III Poisson Processes

Examples, Intuition, and First Definitions

Recall Poisson variables: they are useful for modeling outcomes of a large number of independent trials, where each trial has success probability on the order of $O(\frac{1}{N})$, frequently $\frac{c}{N}$. For instance, $\text{Bin}(N, \frac{c}{N}) \approx \text{Poi}(c)$. Poisson processes operate similarly, with some added spacial or temporal structure.

Eg. 3.1 We consider 1 time dimension. Then, Poisson processes might be well-suited to model arrivals at a Metro self-checkout line; or radioactive decay events; or childbirths. An example which is not well suited might be train arrival times (in a given timestep, the probability of a train arriving is dictated, or influenced, by a schedule; the probability of a train arriving in timestep $n + 1$ is reduced by the event of a train arriving at timestep n).

We consider 2 spacial dimensions. The rainfall on a square meter of pavement is well-modeled (with spacial units one square mm, for instance). We could also consider the locations of shooting stars in a meteor shower.

Some key properties of Poisson processes, which we should keep in mind when modeling:

1. An event at one time or location does not affect the probability of another event at another time or location.
2. Events cannot occur simultaneously or at the same location.

3. Events occur at a rate, i.e. for an event E , $\mathbb{P}[E \in \mathcal{S}] \approx c|\mathcal{S}|$, where $\mathcal{S}$ is some measurable time-space region with "volume" $\mathcal{S}$.

Suppose we have a two-dim region R , which we separate into area ε -sized regions. Suppose the rate of an event (which we will call a "point") in an ε -region is c . Take some subregion $R_1 \subseteq R$ with area A_1 . Then, the total number of ε -regions which contain an event is approximately

$$\approx \text{Bin}\left(\frac{A_1}{\varepsilon}, c\varepsilon\right) \approx \text{Poi}(A_1 \cdot c)$$

If we let $R_2 \subseteq R$ be disjoint from R_1 , with area A_2 , then we similarly model the number of points in A_2 by $\text{Poi}(cA_2)$, which, as a variable, is independent from $\text{Poi}(cA_1)$ above.

For a domain $D \subseteq \mathbb{R}^d$, write $\text{vol}(D)$ for a d -dimensional volume of D . Then, a *homogeneous Poisson process* with rate- c in D is a random set of points $P \subseteq D$, such that, for any

DEF 3.1

$$R_1 \sqcup \dots \sqcup R_k \subseteq D$$

we have the following

- (a) $|P \cap R_1|, \dots, |P \cap R_k|$ are mutually independent.
- (b) $|P \cap R_i| \sim \text{Poi}(c \cdot \text{vol}(R_i))$ for $i \in [k]$.

One Dimensional Processes

When working with one dimension, it is usually time, with points representing occurrences (e.g. births, people entering a store). Let P be a one dimensional rate- c Poisson process. We notate $N_t = |P \cap [0, t]|$, i.e. the number of events occurring up to a time t . Viewed as a stochastic process, we call $(N_t : t \geq 0)$ the *counting process* of P .

DEF 3.2

For $0 \leq s < t$, $N_t - N_s \sim \text{Poi}(c(t - s))$.

PROP 3.1

$$\text{Unwrapping, } N_t - N_s = |P \cap [s, t]| \sim \text{Poi}(c \text{vol}[s, t]) = \text{Poi}(c(t - s)) \quad \square$$

PROOF.

We can form a Poisson process by considering $0 \leq s_1 < t_1 < s_2 < t_2 < \dots < s_n < t_n$, in which case $N_{s_k} - N_{t_k}$ are disjoint.

Let $(N_t : t \geq 0)$ be a counting process. Let $S_k = \inf(t \geq 0 : N_t \geq k)$ be the *arrival time* of the k -th point. Then, we call $(X_k : k \geq 0) : X_k = S_k - S_{k-1}$ the *interarrival times*.

DEF 3.3

DEF 3.4

If $(N_t : t \geq 0)$ is the counting process of a rate- c Poisson process P on $[0, \infty)$, then the interarrival times $(X_k : k \geq 1)$ are independent and $\text{Exp}(c)$ distributed.

PROP 3.2

$$\text{Recall that } \text{Poi}(\alpha) \text{ has density } \frac{\alpha^k e^{-\alpha}}{k!} \text{ and } \text{Exp}(c) \text{ has density } ce^{-ct}. \quad \square$$

PROOF.

We have a converse to this: if $(E_k : k \geq 1)$ are independent and $\text{Exp}(c)$ distributed, then $(P_k : P_k = E_1 + \dots + E_k : k \geq 1)$ is a rate- c Poisson process on $[0, \infty)$.

PROP 3.3

E.g. length for $d = 1$
or area for $d = 2$
"points" = "events"

By homogeneous, we
refer to the fact that
the rate is constant
over regions

Generally speaking,
any process which
satisfies $N_0 \geq 0$ and
 $N_s \leq N_t \leq \lim_{r \rightarrow t} N_r + 1$
for $0 \leq s \leq t$ is a
counting process

i.e. the amount of
time that passes
between the $k - 1$
and k -th points

Eg. 3.2 Given a self-checkout counter which averages 1 arrival per minute around 2PM on a Monday, we ask for the probability that 8 people arrive between 2:00-2:06 and 3 people arrive between 2:06-2:10. With the notation above, this is

$$\begin{aligned}\mathbb{P}(N_6 - N_0 = 8, N_6 - N_{10} = 3) &= \mathbb{P}(|P \cap [0, 6]| = 8, |P \cap (6, 10]| = 3) \\ &= \mathbb{P}(|P \cap [0, 6]| = 8) \mathbb{P}(|P \cap (6, 10]| = 3) \\ &= \mathbb{P}(\text{Poi}(6) = 8) \mathbb{P}(\text{Poi}(4) = 3) \\ &= \frac{6^8 e^{-6}}{8!} \frac{4^3 e^{-4}}{3!} \approx 0.02\end{aligned}$$

Eg. 3.3 Consider a sky of shooting stars, where we observe 1 star per two minutes. In minute units, $c = \frac{1}{2}$. Then we ask for the probability of seeing 4 stars in the first 10 minutes and 20 stars in the first 30 minutes. This is

$$\mathbb{P}(N_{10} = 4, N_{30} = 20) = \mathbb{P}(N_{10} = 4, N_{30} = N_{10} + 16)$$

With disjointed intervals, we can now write this as

$$\mathbb{P}(|P \cap [0, 10]| = 4, |P \cap (10, 30]| = 16) = \mathbb{P}(\text{Poi}(5) = 4) \mathbb{P}(\text{Poi}(10) = 16)$$

Eg. 3.4 We'll consider a multidimensional example: raindrop locations on a square of sidewalk. Suppose that we observe a mean of λ raindrops per square centimeter. Then the probability of seeing k raindrops in a region L is $\text{Poi}(\lambda \text{Area}(L)) = k$

Variable-Rate Processes

We'll begin with an example. Suppose the rate of shooting stars during a meteor shower changes day by day via λ . We ask how many shooting stars fall during a time interval $[s, t]$. When considering a small time interval $[s, s + \delta] \subseteq [s, t]$, the rate can be approximated as constant. By stitching together many small disjoint intervals, we can reconstruct the entire interval, i.e., letting $r = t - s$

$$\mathbb{P}(|P \cap [s, t]| = k) \approx \mathbb{P}\left(\left(\sum_{i=1}^N \text{Poi}\left(\frac{r}{N} \lambda\left(s + \frac{ri}{N}\right)\right)\right) = k\right)$$

This motivates the following definition:

DEF 3.5 Given a domain $D \subseteq \mathbb{R}^d$ and a function $\lambda : D \rightarrow [0, \infty)$, a *Poisson process* with rate- λ in D is a random set of points $P \subseteq D$ such that, for any

$$R_1 \sqcup \cdots \sqcup R_k \subseteq D$$

we have the following:

- (a) $|P \cap R_1|, \dots, |P \cap R_k|$ are mutually independent.

$$(b) |P \cap R_i| \sim \text{Poi}\left(\int_{D_i} \lambda(\vec{x}) dA\right) \text{ for } i \in [k].$$

Thinning and Superposition

THINNING

Thinning a Poisson process is akin to restricting our attention to specific "types" of points.

Let P be Poisson process in domain D , and suppose that each point of P has a *type*, defined to be a labeling $P \rightarrow [k]$ for some $k \in \mathbb{Z}_{>0}$. We define a probability measure on type assignments, i.e. $\mathbb{P}(\text{point has type } i) = q_i$, where $\sum_{i \in [k]} q_i = 1$ and $q_i \geq 0$. We suppose that type assignments are independent.

DEF 3.6

Write $P_i = \{p \in P : p \text{ has type } i\}$. Then $P_1, \dots, P_k$ are independent Poisson processes. In particular, if P has rate- λ , then P_i has rate- λq_i .

PROP 3.4

We will only give a proof sketch. We must show that, for any disjoint regions $R_1, \dots, R_n \subseteq D$ and non-negative integers $(m_{i,j}) : i \in [k], j \in [n]$, then

PROOF.

$$\begin{aligned} \mathbb{P}\left(\bigcap_{i \in [k]} \bigcap_{j \in [n]} |P_i \cap R_j| = m_{i,j}\right) &= \prod_{i \in [k]} \prod_{j \in [n]} \mathbb{P}\left(|P_i \cap R_j| = m_{i,j}\right) \\ &= \prod_{i \in [k]} \prod_{j \in [n]} \mathbb{P}\left(\text{Poi}(q_i \lambda \text{vol}(R_j)) = m_{i,j}\right) \end{aligned}$$

in the case of homogeneous processes. With variable rate, $\lambda \text{vol}(R_i)$ becomes $\int_{R_i} \lambda(\vec{x}) dx$.

We will not prove this, in generatlity, but will provide the calculation for $n = 2$ regions and $k = 2$ types. Fix regions R_1, R_2 with volumes v_1, v_2 , respectively. We'll write $q_2 = 1 - q_1$. Then, for $r_1, r_2, b_1, b_2 \geq 0$ (think type red and type blue in regions R_1 and R_2), the probability above is given by

$$\begin{aligned} &= \mathbb{P}(|P_r \cap R_1| = r_1, |P_b \cap R_1| = b_1, |P_r \cap R_2| = r_2, |P_b \cap R_2| = b_2) \\ &= \mathbb{P}(|P \cap R_1| = r_1 + b_1, |P \cap R_2| = r_2 + b_2) \mathbb{P}(\text{Bin}(r_1 + b_1, q_1) = r_1) \mathbb{P}(\text{Bin}(r_2 + b_2, q_1) = r_2) \\ &= \mathbb{P}(\text{Poi}(\lambda v_1) = r_1 + b_1) \cdot \mathbb{P}(\text{Poi}(\lambda v_2) = r_2 + b_2) \cdot \binom{r_1 + b_1}{r_1} q_1^{r_1} q_2^{b_1} \binom{r_2 + b_2}{r_2} q_1^{r_2} q_2^{b_2} \\ &= \frac{e^{-\lambda v_1} (\lambda v_1)^{r_1 + b_1}}{(r_1 + b_1)!} \frac{(r_1 + b_1)!}{r_1! b_1!} \frac{e^{-\lambda v_2} (\lambda v_2)^{r_2 + b_2}}{(r_2 + b_2)!} \frac{(r_2 + b_2)!}{r_2! b_2!} q_1^{r_1 + r_2} q_2^{b_1 + b_2} \\ &= \frac{e^{-\lambda v_1} (\lambda v_1 q_1)^{r_1}}{r_1!} \frac{e^{-\lambda v_1} (\lambda v_1 q_2)^{b_1}}{b_1!} \frac{e^{-\lambda v_2} (\lambda v_2 q_1)^{r_2}}{r_2!} \frac{e^{-\lambda v_2} (\lambda v_2 q_2)^{b_2}}{b_2!} \end{aligned}$$

which is what we were after. For a general k , we do a similar argument with a multinomial distribution rather than binomial distributions. (Induction could also work.) $\square$

SUPERPOSITION

Superposition is akin to joining two processes. One could "unthin" a thinned process this way.

PROP 3.5 Let P be a rate- λ Poisson process in a domain D and Q be a rate- β process, independent from P , also on D . Then $P \cup Q$ is a rate- $(\lambda + \beta)$ Poisson process in D .

PROOF.

Again, we provide a sketch (two processes, one region). For any region $R \subseteq D$, we have

$$\begin{aligned}
 \mathbb{P}(|(P \cup Q) \cap R| = k) &= \sum_{i=0}^k \mathbb{P}(|P \cap R| = i, |Q \cap R| = k - i) \\
 &= \sum_{i=0}^k \mathbb{P}(\text{Poi}(\lambda \text{vol}(R)) = i) \mathbb{P}(\text{Poi}(\beta \text{vol}(R)) = k - i) \\
 &= \sum_{i=0}^k \frac{e^{-\lambda \text{vol}(R)} (\lambda \text{vol}(R))^i}{i!} \frac{e^{-\beta \text{vol}(R)} (\beta \text{vol}(R))^{k-i}}{(k-i)!} \\
 &= \frac{e^{-(\lambda+\beta) \text{vol}(R)} \text{vol}(R)^k}{k!} \sum_{i=0}^k \lambda^i \beta^{k-i} \frac{k!}{i!(k-i)!} \\
 &= \frac{e^{-(\lambda+\beta) \text{vol}(R)} \text{vol}(R)^k}{k!} (\lambda + \beta)^k = \mathbb{P}(\text{Poi}((\lambda + \beta) \text{vol}(R)) = k)
 \end{aligned}$$

by binomial theorem. Really, all we did is prove that the sum of Poisson variables is Poisson with rate equal to the sum of rates of the summands. $\square$

It is a fact* (not too hard to prove) that $\min(P, Q) \sim \text{Exp}(\lambda + \beta)$, where P, Q are independent $\text{Exp}(\lambda)$ and $\text{Exp}(\beta)$, respectively. Consider the arrival time of the first point in P , which by [Prop 3.2](#) is $\text{Exp}(\lambda)$ distributed, and similarly for Q , which is $\text{Exp}(\beta)$ distributed. The result we proved above suggests that the arrival time of the first point in $P \cup Q$ is $\text{Exp}(\lambda + \beta)$ distributed. But this is the minimum of arrivals of the first point in P and Q , so our fact* is consistent.

We've reverted to processes as "rate- λ ." Why rate? One can show that, for a one-dimensional process, the counting process $N(t)$ satisfies $\frac{d}{dt} \mathbb{E}[N(t)] = \lambda$ at $t = 0$ (and, in fact, for all $t \geq 0$). Thus, we accumulate points at roughly rate- λ per time unit.

IV Continuous-Time Markov Chains

It will be useful to observe that $\mathbb{P}(\text{Exp}(\lambda) \leq h) \approx \lambda h + O(h^2)$.

We'll piggy back off the previous topic. Consider a supermarket checkout queue length as a function of time, $L_t : t \geq 0$. We will model the arrivals and departures as independent rate- α and rate- δ Poisson processes, respectively. Instead of encoding L_t in the language of Poisson processes (which we certainly could do), we can model it as a Markov chain.

Excuse this notation: it does not refer to $P \cap [0, t]$, as in the previous unit

We consider states $V = \mathbb{Z}_{\geq 0}$. Let $E^{(\alpha)}$ and $E^{(\delta)}$ be independent exponential random variables with rate α and δ , respectively. Roughly (but not literally), these encode the interarrival and interdeparture times. In particular, we follow the rule that

$$L_{t+\text{interarrival time}} = \begin{cases} i+1 & E^{(\alpha)} \leq E^{(\delta)} \\ i-1 & E^{(\alpha)} > E^{(\delta)} \end{cases} \quad \text{given } L_t = i$$

As a transition:

$$\mathbb{P}_i(L_{\text{first change}} = i+1) = \mathbb{P}(E^{(\alpha)} \leq E^{(\delta)}) = \mathbb{P}(\min(E^{(\alpha)}, E^{(\delta)}) = E^{(\alpha)}) = \frac{\alpha}{\alpha + \delta}$$

We also observe that the transition time between queue changes is $\min(E^{(\alpha)}, E^{(\delta)})$ distributed. One derives from basic probability that this is $\text{Exp}(\alpha + \delta)$ -distributed.

The *transition function* of a continuous-time Markov chain is $P(t) := (P_{ij}(t))_{i,j \geq 0}$ where

DEF 4.1

$$P_{ij}(t) = \mathbb{P}_i(L_t = j) = \mathbb{P}(L_t = j | L_0 = i)$$

Note that this is a different notion than the "probability that the next transition from i occurs in time t and goes to j ." Instead, this is the "probability that, from i , we will be at state j time t into the future." $P(t)$, as a quantity, is difficult to evaluate. We will consider a simpler object.

We call Q_{ij} the *generator matrix*, or the infinitesimal generator, which is given by

DEF 4.2

$$Q_{ij} = \lim_{h \rightarrow 0} \frac{\mathbb{P}_i(L_h = j) - \mathbb{P}_i(L_0 = j)}{h} = \lim_{h \rightarrow 0} \frac{P_{ij}(h) - P_{ij}(0)}{h} = P'_{ij}(0)$$

i.e. the derivatives of $P(0)$.

Fig. 4.1 We consider the checkout queue example from above. For $i > 0$, we have

$$\mathbb{P}_i(L_h = i) = \mathbb{P}(E^{(\alpha)} > h, E^{(\delta)} > h) + O(h^2) = e^{-(\alpha+\delta)h} + O(h^2)$$

Then, as $h \rightarrow 0$, we have

$$\frac{\mathbb{P}_i(L_h = i) - \mathbb{P}_i(L_0 = i)}{h} \rightarrow \frac{e^{-(\alpha+\delta)h} - 1}{h} \rightarrow -(\alpha + \delta)$$

so $Q_{i,i} = -(\alpha + \delta)$. Similarly, for $i > 0$, we have

$$\begin{aligned} \mathbb{P}_i(L_h = i+1) &= \mathbb{P}_i(E^{(\alpha)} \leq h < E^{(\delta)}) + O(h^3) \\ &= (1 - e^{-\alpha h})e^{-\delta h} + O(h^3) = \alpha h + O(h^2) \end{aligned}$$

We can conclude from this that $Q_{i,i+1} = \alpha$. Similar arguments show that $Q_{0,0} = -\alpha$, $Q_{0,1} = \alpha$, and $Q_{i,i-1} = \delta$ for $i > 0$. Thus, the generator matrix for P is

$$\begin{pmatrix} -\alpha & \alpha & 0 & 0 & 0 & \cdots \\ \delta & -\alpha - \delta & \alpha & 0 & 0 & \cdots \\ 0 & \delta & -\alpha - \delta & \alpha & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Eg. 4.2 Consider a generator matrix

DEF 4.3 A *continuous-time Markov chain* is a state space V with a stochastic process $(X_t : t \geq 0)$ such that

(a) $X_t \in V \forall t \geq 0$

(b) For all $s, t \geq 0$,

$$\mathbb{P}(X_{t+s} = v | X_r = x_r : r \in [0, t]) = \mathbb{P}(X_{t+s} = v | X_t = x_t)$$

DEF 4.4 A chain is called *continuous time-homogenous* if, additionally,

$$\mathbb{P}(X_{t+s} = v | X_r = x_r : r \in [0, t]) = \mathbb{P}(X_s = v | X_0 = x_t)$$

From now all, all chains in this unit will be continuous time-homogeneous. In particular, we will focus on chains with finitely many transitions on a finite timescale.

Let X_t be a CTMC with state space V . Let $T_1 = \inf(t \geq 0 : X_t \neq X_0)$ and $T_2 = \inf(t \geq T_1 : X_t \neq X_{T_1})$. Generally,

$$T_n = \inf(t \geq T_{n-1} : X_t \neq X_{T_{n-1}})$$

DEF 4.5 we call these the *jump times* of the chain.

DEF 4.6 If $\mathbb{P}_v(T_k \text{ diverges}) = 1$, we say that the chain is *non-explosive*. Otherwise, the chain is called

DEF 4.7 *explosive*. For an explosive chain, we see that we will observe infinitely many transitions in a finite time interval.

DEF 4.8 The *embedded chain* of X_t , denoted $(\tilde{X}_i : i \in \mathbb{Z}_{\geq 0})$, is defined by

$$\tilde{X}_i := X_{T_i}$$

Note that the embedded chain is a discrete Markov chain, which has transition matrix

$$\tilde{P} := (\tilde{P}_{uv})_{u,v \in V} \quad \text{where} \quad \tilde{P}_{uv} = \mathbb{P}(X_{T_1} = v | X_0 = u)$$

with $\tilde{P}_{u,u} = 0$, since $X_{T_1} \neq X_0$ by definition.

PROP 4.1 Conditional on $X_0 = v$, T_1 is exponentially distributed.

PROOF.

To show this, we prove that T_1 has the memoryless property, i.e. that

$$\mathbb{P}_v(T_1 > s + t | T_1 > t) = \mathbb{P}_v(T_1 > s)$$

It is a fact of probability that the only continuous memoryless variable is exponential. $\square$

DEF 4.9 Oftentimes we call $(T_1 | X_0 = v)$ the *holding time*, and we write $(T_1 | X_0 = v) \sim \text{Exp}(q_v)$. The values $(q_v : v \in V)$ and the embedded chain $\tilde{P}$ fully describe the Markov chain.

Theorem 4.1 Continuous-Time Markov Characterization

Let X_t be a continuous-time Markov chain. Let $(q_v : v \in V)$ be such that $(T_1 | X_0 = v) \sim \text{Exp}(q_v)$ and let $(\tilde{P}_{uv} : u, v \in V)$ be the transition matrix of the embedded chain. Then $(X_t : t \geq 0)$ can be reconstructed as follows:

1. Suppose at time t , we are at a state $X_t = u$.
2. We will remain at this state for a duration distributed by $\text{Exp}(q_u)$.
3. The probability of advancing to state v is given by $\tilde{P}_{u,v}$.

Fix u . For $v \neq u$, let $E_{u,v} \sim \text{Exp}(q_u \tilde{P}_{u,v})$. Suppose $X_t = u$. Then, let $E^* = \min_{v \neq u \in V} E_{u,v}$ and $v^* = \text{argmin}_{v \neq u \in V} E_{u,v}$. Then, X_t will advance to a state according to the distribution of v^* , and it will do so in a time duration according to the distribution of E .

PROP 4.2

To verify this claim, we need to prove that $\mathbb{P}_u(X_{T_1} = v) = \tilde{P}_{u,v}$ and that the duration of time spent at state u is distributed according to $\text{Exp}(q_u)$.

PROOF.

We have

$$E^* = \min_{v \neq u} E_{u,v} \sim \text{Exp}\left(\sum_{v \neq u \in V} q_u \tilde{P}_{u,v}\right) = \text{Exp}\left(q_u \sum_{v \neq u \in V} \tilde{P}_{u,v}\right) = \text{Exp}(q_u)$$

Similarly,

$$\mathbb{P}_u(X_{T_1} = v) = \mathbb{P}(E_{u,v} = \min_{v \neq u} E_{u,v}) = \frac{q_u \tilde{P}_{u,v}}{\sum_{w \neq u \in V} q_u \tilde{P}_{u,w}} = \tilde{P}_{u,v}$$

□

This view of a continuous-time Markov chain is called the *exponential alarm clock*. Below, we prove that the times $E_{u,v}$ are distributed according to $\text{Exp}(Q_{u,v})$, where we recall the generator matrix Q .

DEF 4.10

Recall the generator of a continuous-time chain, given by $Q_{u,v} = P'_{u,v}(0)$. We can write

PROP 4.3

$$Q_{u,v} = \begin{cases} q_u \tilde{P}_{u,v} & v \neq u \\ -q_u & v = u \end{cases}$$

From this, we conclude that a continuous-time Markov chain is characterized entirely by the generator matrix Q (and that the sum of row entries is always 0). From Q , we can entertain either the alarm clock view directly, or the wait-embedded-chain view by reconstructing $\tilde{P}$.

For infinitesimal h , $\mathbb{P}_u(X_h = v) = \mathbb{P}(E \leq h, E = E_{u,v}) + O(h^2)$. When $v \neq u$, then,

PROOF.

$$\begin{aligned} \mathbb{P}_u(X_h = v) &= \mathbb{P}(E \leq h) \mathbb{P}(E = E_{u,v} | E \leq h) + O(h^2) \\ &= [q_u h + O(h^2)] [\tilde{P}_{u,v}] + O(h^2) = q_u h \tilde{P}_{u,v} + O(h^2) \end{aligned}$$

For diagonal terms, we derive similarly that $\mathbb{P}_u(X_h = v) = -q_u h + O(h^2)$. The definition of the derivative gives the result. $\square$

Eg. 4.3 Let P be a rate- λ Poisson process on $[0, \infty)$. Recall the arrival times $S_k : k \geq 0$. Let $D_k : k \geq 0$ be IID integer random variables, with $p_i := \mathbb{P}(D_1 = i)$. Then we set

$$X_t := \sum_{i=1}^k D_i \quad \text{for any} \quad S_k \leq t \leq S_{k+1}$$

Previously, we defined Q , the generator matrix, in terms of P . Can we recover P from a given Q ? Recall that

$$P_{uv}(t) = \mathbb{P}_u(X_t = v) \quad \text{and} \quad Q_{uv} = P'_{uv}(0)$$

The answer to this question is yes, but we will require a continuous-time version of the Chapman-Kolmogorov equations.

Theorem 4.2 Chapman-Kolmogorov for CTMC

For all $s, t \geq 0$, we have $P(s+t) = P(s)P(t)$

PROOF.

$$\begin{aligned} P_{u,v}(s+t) &= \mathbb{P}_u(X_{s+t} = v) = \sum_{w \in V} \mathbb{P}_u(X_s = w, X_{s+t} = v) \\ &= \sum_{w \in V} \mathbb{P}_u(X_s = w) \mathbb{P}_u(X_{s+t} = v | X_s = w) \end{aligned}$$

Then, by the Markov property for continuous chains,

$$\sum_{w \in V} \mathbb{P}_u(X_s = w) \mathbb{P}_u(X_{s+t} = v | X_s = w) = \sum_{w \in V} \mathbb{P}_u(X_s = w) \mathbb{P}_w(X_t = v) = (P(s)P(t))_{u,v}$$

as desired. $\square$

PROP 4.4 If $P(t)$ has generator Q , then $P'(t) = P(t)Q$ and $P'(t) = QP(t)$, which are called Kolmogorov's *forward and backward equations*, respectively.

PROOF.

We first note that $Q_{u,v}$ can be written as

$$Q_{u,v} = \lim_{h \rightarrow 0} \frac{P_{u,v}(h) - P_{u,v}(0)}{h} = \lim_{h \rightarrow 0} \frac{P_{u,v}(h) - I_{u,v}}{h}$$

We have

$$\frac{P(t+h) - P(t)}{h} = \frac{P(t)P(h) - P(t)}{h} = \frac{P(t)[P(h) - I]}{h}$$

Then, for the limit,

$$\lim_{h \rightarrow 0} \frac{P(t)[P(h) - I]}{h} = P(t) \lim_{h \rightarrow 0} \frac{P(h) - I}{h} = P(t)Q$$

An identical argument, using $P(t + h) = P(h)P(t)$ instead of $P(t)P(h)$, provides the backward equation. $\square$

For a continuous-time Markov chain, we can write

PROP 4.5

$$P(t) = e^{tQ} = \sum_{k=0}^{\infty} \frac{(tQ)^k}{k!}$$

If $f : [0, \infty) \rightarrow \mathbb{R}$ satisfies $f(0) = 1$ and $f'(x) = qf(x)$, then we can easily recover $f(x) = e^{qx}$. In our setting, $P(0) = I$ and $P'(t) = QP(t)$, so we conclude that $P(t) = e^{tQ} = \sum_{k=0}^{\infty} \frac{(tQ)^k}{k!}$. $\square$

PROOF.

Eg. 4.4 Let $(X_t) : t \geq 0$ be the counting process of a rate- λ Poisson process. Then

$$P_{ij}(t) = \mathbb{P}(\text{Poi}(\lambda t) = j - i) = \begin{cases} 0 & j < i \\ \frac{e^{-\lambda t}(\lambda t)^{j-i}}{(j-i)!} & j \geq i \end{cases}$$

In this case, $Q_{i,i} = -\lambda$ and $Q_{i,i+1} = \lambda$, and 0 elsewhere. For $i \geq 0$, using the backward equation, we have

$$P'_{i,i}(t) = (QP(t))_{i,i} = \sum_{k \geq 0} Q_{ik}P_{ki}(t) = Q_{ii}P_{ii}(t) = -\lambda e^{-\lambda t} = -\lambda P_{ii}(t)$$

One comes to the same result using the forward equation.

Now, for $j > i$, using the backward equation

$$P'_{i,j}(t) = (QP(t))_{i,j} = \sum_{k \geq 0} Q_{ik}P_{kj}(t) = -\lambda P_{i,j}(t) + \lambda P_{i+1,j}(t)$$

Using the forward equation,

$$P'_{i,j}(t) = (P(t)Q)_{i,j} = \sum_{k \geq 0} P_{i,k}(t)Q_{k,j} = \lambda P_{i,j-1}(t) - \lambda P_{i,j}(t)$$

So we conclude that $P_{i+1,j}(t) = P_{i,j-1}(t)$. Then, solving the resulting differential equation gives

$$P_{i,j}(t) = \frac{e^{-\lambda t}(\lambda t)^{j-i}}{(j-i)!}$$

Fig. 4.5 Consider a two-state chain $1 \leftrightarrow 2$, where $\mathbb{P}(1 \mapsto 2) = \lambda$ and $\mathbb{P}(2 \mapsto 1) = \mu$. Then, the Q matrix is

$$Q = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}$$

Observe, $P_{1,1}(t) = 1 - P_{1,2}(t)$ and $P_{2,2}(t) = 1 - P_{2,1}(t)$. Also note that $P_{1,1}(0) = P_{2,2}(0) = 1$. Then,

$$P'_{1,1}(t) = (P(t)Q)_{1,1} = -\lambda P_{1,1}(t) + \mu P_{1,2}(t) = -\lambda P_{1,1}(t) + \mu(1 - P_{1,1}(t))$$

Then, simplifying, this gives $P'_{1,1}(t) = \mu - (\lambda + \mu)P_{1,1}(t)$, so, solving the differential equation yields

$$P_{1,1}(t) = \frac{\mu}{\lambda + \mu} + \frac{\lambda}{\lambda + \mu} e^{-(\lambda + \mu)t}$$

Similarly, we can conclude that

$$P_{2,2}(t) = \frac{\lambda}{\lambda + \mu} + \frac{\mu}{\lambda + \mu} e^{-(\lambda + \mu)t}$$

Taking limits, we find, nicely, that

$$P(t) \rightarrow \begin{pmatrix} \frac{\mu}{\lambda + \mu} & \frac{\lambda}{\lambda + \mu} \\ \frac{\mu}{\lambda + \mu} & \frac{\lambda}{\lambda + \mu} \end{pmatrix}$$

For the same two-state chain, then,

$$\tilde{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

so the jump chain does not converge.

Given a continuous-time Markov chain with generator Q and state space V , we say that $u \rightarrow v$, or that v is *accessible* from u , if there exists some $k \geq 0$ and states $u = u_0, u_1, \dots, u_k = v$ such that $Q_{u_i, u_{i+1}} > 0$ for $i \in [0, k)$. By extension, we call the chain *irreducible* if $\forall u, v \in V$, $u \rightarrow v$.

For a non-explosive continuous-time Markov chain with finite state space V , generator Q , and states $u, v \in V$, the following are equivalent:

1. $u \rightarrow v$
2. $\exists t > 0 : P_{u,v}(t) > 0$
3. $\forall t > 0, P_{u,v}(t) > 0$

PROOF.

3 $\implies$ 2 is immediate. For 2 $\implies$ 1, fix $t > 0$ with $P_{u,v}(t) > 0$. Then, since $P(t) = e^{tQ} = \sum_{k \geq 0} \frac{t^k}{k!} Q^k$, there must be some $k \geq 0$ with $Q^k_{u,v} > 0$. Then, we must find a positive probability path of length k from u to v . For 1 $\implies$ 3, suppose $Q^k_{u,v} > 0$ for

some $k \geq 0$. Then, for any $t > 0$, $P_{u,v}(t) \geq \frac{t^k}{k!} Q_{u,v}^k > 0$.

We conclude that $1 \iff 2$, so also $1 \iff 3$, and so $2 \iff 3$ follows. While this proof is symbolic, it (sh/c)ould be re-thought of probabilistically. $\square$

STATIONARY AND LIMITING DISTRIBUTIONS

Let $P(t)$ be the transition function of a continuous-time Markov chain with state space V . We say that a probability distribution π on V is *stationary* if

DEF 4.14

$$\pi P(t) = \pi \quad \forall t \geq 0$$

Similarly, we say that π is *limiting* if, for any $u, v \in V$,

DEF 4.15

$$\lim_{t \rightarrow \infty} P_{u,v}(t) = \pi_v$$

Just as with discrete chains, for an initial distribution α , we have that $\alpha P(t) \rightarrow \pi$.

Eg. 4.6 Suppose that the transition matrix $\tilde{P}$ for the embedded chain is regular, such that it has a unique stationary, limiting distribution π . Then π_v is long-run proportion of time that the embedded chain spends at state v (but not necessarily the continuous-time chain). Each time the continuous-time chain visits v , we remain there for, on average, time $\frac{1}{q_v}$, where $(T_1 | X_0 = v) \sim \text{Exp}(q_v)$. This suggests that, in the long run, a stationary distribution ψ for the continuous-time chain will have

$$\psi_v \propto \frac{\pi_v}{q_v} \implies \psi_v = \frac{\pi_v/q_v}{\sum_{u \in V} \pi_u/q_u}$$

Let X_t be a continuous-time Markov chain with generator Q . Then π is a stationary distribution for the continuous-time chain if and only if $\pi Q = \vec{0}$. Equivalently,

PROP 4.7

$$\sum_{u \neq v \in V} \pi_u Q_{u,v} = \pi_v q_v \quad \forall v \in V$$

We call this last characterization the *global balance equation*.

DEF 4.16

Recall two characterizations of Q : for one, $Q = P'(0)$ and, for two, $P'(t) = QP(t)$ (i.e. the backward equation). Suppose that π is stationary, such that $\pi P(t) = \pi$. Then $\frac{d}{dt}[\pi P(t)] = \vec{0}$, i.e. $\pi P'(t) = \vec{0}$. But then $\pi Q = \pi P'(0) = \vec{0}$.

PROOF.

Conversely, suppose that $\pi Q = \vec{0}$. Then, $\frac{d}{dt}[\pi P(t)] = \pi P'(t) = \pi QP(t) = \vec{0}P(t) = \vec{0}$. Hence, $\pi P(t) = \pi P(0) \quad \forall t \geq 0$, since we observe no rate of change. Since $P(0) = I$, $\pi P(t) = \pi$ as desired.

For the second characterization, we have, from $\pi Q = \vec{0}$,

$$\sum_{u \in V} \pi_u Q_{u,v} = 0 \quad \forall v \in V$$

Since $Q_{v,v} = -q_v$ we have that

$$\sum_{u \neq v \in V} \pi_u Q_{u,v} = -\pi_v Q_{v,v} = \pi_v q_v$$

□

The global balance equation, heuristically, says that, with states distributed according to π , the time-rate at which we enter v from other states (in the "alarm clock view") is equivalent to the time-rate at which we exit v once there.

DEF 4.17 Similarly, the *local balance equation* is the property that

$$\forall u, v \in V, \pi_v Q_{v,u} = \pi_u Q_{u,v}$$

PROP 4.8 If π and Q satisfy local balance, then they satisfy global balance. Hence, in this case, π is a stationary distribution for the continuous-time chain. The converse need not hold.

DEF 4.18 If π and Q satisfy local balance, then we call the chain *time-reversible*.

DEF 4.19 The *undirected transition graph* of a continuous-time Markov chain with generator Q and state space V has vertex set V and edge set $E = \{(u, v) : Q_{u,v} > 0 \text{ or } Q_{v,u} > 0\}$. Similarly,

DEF 4.20 the *directed transition graph* consists of edges $E = \{(u, v) : Q_{u,v} > 0\}$.

Theorem 4.3 Criterion for Continuous-Time Reversibility

If the undirected transition graph of an irreducible, non-explosive continuous-time Markov chain is a tree, then it is time-reversible.

PROP 4.9 A limiting distribution is also a stationary distribution for a continuous-time Markov chain.

PROOF.

Fix an initial distribution α . For any $c > 0$, we have

$$\lambda P(c) = \left(\lim_{t \rightarrow \infty} \alpha P(t) \right) P(c) = \lim_{t \rightarrow \infty} \alpha P(t + c) = \lambda$$

using Chapman-Kolmogorov ([Thm 4.2](#)).

□

Fundamental Theorem of Continuous-Time Markov Chains

Theorem 4.4 Fundamental Theorem of Continuous-Time Markov Chains

Let X_t be a continuous-time Markov chain with finite state space V and transition function $P(t)$. If it is irreducible, then there is a unique stationary distribution π which is also limiting.

Consider the matrix $\hat{P}$, where $\hat{P}_{u,v} := \mathbb{P}_u(X_1 = v)$. Note that this is distinct from $\tilde{P}$: we are considering the state of the chain at integer times only, instead of the jumps themselves. This discretization, $\hat{P}$, is also a discrete Markov chain, equal to $(X_n : n \in \mathbb{Z}_{\geq 0})$. Let $P^{(h)}$

Positive integers are not special: we could consider times $X_{n\pi}$ for $n \geq 0$, say.

be the discretization provided by $(X_{nh} : n \in \mathbb{Z}_{\geq 0})$. Observe that a stationary distribution for a "fine" discretization will remain stationary for a "coarser" discretization. With this thought in mind, we turn to the proof of the theorem.

PROOF.

Consider the discretization $(X_n : n \in \mathbb{Z}_{\geq 0})$. This has transition matrix $P(1)$. Since the continuous-time chain X_t is irreducible, $P(1)$ is positive for all entries. We conclude that $P(1)$ is regular, so the discretization has a unique limiting, stationary distribution λ . This is to say, $\lambda P(1) = \lambda$.

Now, fix $\varepsilon > 0$. Let $h > 0$ be small enough that $\mathbb{P}_x(T_1 < h) < \frac{\varepsilon}{2}$ for all states $x \in V$. We can then choose h such that $\frac{1}{h} \in \mathbb{Z}_{\geq 0}$.

Consider the discrete chain $Y := (X_{nh} : n \in \mathbb{Z}_{\geq 0})$, which has transition matrix $\hat{P} = P(h)$. Once again, Y_n is regular, and thus has a limiting distribution. But we have

$$\lim_{n \rightarrow \infty} \mathbb{P}_u(Y_n = v) = \lim_{n \rightarrow \infty} \mathbb{P}_u(X_n = v) = \lambda_v$$

We conclude that the Y chain also has limiting distribution λ . Thus, fix N s.t.

$$\forall n \geq N, u, v \in V \quad |\mathbb{P}_u(Y_n = v) - \lambda_v| < \frac{\varepsilon}{2}$$

Thus, for any $n \geq N$, $nh \leq t < (n+1)h$, we have

$$\begin{aligned} P_{u,v}(t) &= \mathbb{P}_u(X_t = v) = \mathbb{P}_u(X_{nh} = v, X_t = v) + \mathbb{P}_u(X_{nh} \neq v, X_t = v) \\ &= \mathbb{P}_u(X_{nh} = v, X_t = v) + \sum_{w \neq v \in V} \mathbb{P}_u(X_{nh} = w, X_t = v) \\ &= \mathbb{P}_u(X_{nh} = v, X_t = v) + \sum_{w \neq v \in V} \mathbb{P}_u(X_t = v | X_{nh} = w) \mathbb{P}_u(X_{nh} = w) \\ &\leq \mathbb{P}_u(X_{nh} = v, X_t = v) + \sum_{w \neq v \in V} \mathbb{P}_w(T_1 < h) \mathbb{P}_u(X_{nh} = w) \\ &< \mathbb{P}_u(X_{nh} = v, X_t = v) + \sum_{w \neq v \in V} \frac{\varepsilon}{2} \mathbb{P}_u(X_{nh} = w) \\ &\leq \mathbb{P}_u(X_{nh} = v, X_t = v) + \frac{\varepsilon}{2} \end{aligned}$$

As for the other summand, we have

$$P_u(X_{nh} = v, X_t = v) = \mathbb{P}_u(X_{nh} = v) - \mathbb{P}_u(X_{nh} = v, X_t \neq v) > \mathbb{P}_u(X_{nh} = v) - \frac{\varepsilon}{2}$$

Thus, we an upper

$$P_{u,v}(t) \leq \mathbb{P}_u(X_{nh} = v) + \frac{\varepsilon}{2} = \mathbb{P}_u(Y_n = v) + \frac{\varepsilon}{2} \leq \lambda_v + \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \lambda_v + \varepsilon$$

and lower

$$P_{u,v}(t) \geq \mathbb{P}_u(X_{nh} = v, X_t = v) > \mathbb{P}_u(X_{nh} = v) - \frac{\varepsilon}{2} = \mathbb{P}_u(Y_n = v) - \frac{\varepsilon}{2} < \lambda_v - \frac{\varepsilon}{2} - \frac{\varepsilon}{2} = \lambda_v - \varepsilon$$

i.e. we find a time that is a rare enough duration

bound. We deduce that $|P_{u,v}(t) - \lambda_v| < \varepsilon$ when $nh \leq t < (n+1)h$ and n large enough. We conclude that $|P_{u,v}(t) - \lambda_v| < \varepsilon$ for any $t \geq Nh$. Therefore, $|P_{u,v}(t) - \lambda_v| \rightarrow 0$, as desired. We refer to [Prop 4.8](#) to show that this is also stationary. $\square$

YULE PROCESSES

We will describe a continuous-time variant of branching processes, seen in Part II. Whereas, previously, we simply considered the number of individuals in an integer-valued generation (those in a given generation spend exactly 1-generation worth of time alive), we suppose that the lifetimes of individuals are modeled by exponential random variables. In short, the separation of generations breaks down.

Suppose an individuals' lifespan is distributed according to $\text{Exp}(1)$, and suppose, at death, an individual has 2 children with probability 1. If we ignore the time aspect, this underlying process will be a binary tree.

Let T_∞ be the infinite, complete, rooted binary tree. For any vertex v , we'll denote by $p(v)$ its parent, and we say $e < v$ if the root-to- v path contains the edge e . For each edge $e \in E$, we associate a variable $X_e \sim \text{Exp}(1)$ representing individual lifetimes.

With this notation, an individual $v \in V$ lives

$$\text{from time } A(v) = \sum_{e < p(v)} X_e \text{ to time } B(v) = \sum_{e < v} X_e$$

At a given time $t \geq 0$, we can work out the number of individuals alive,

$$Y(t) = \#\{v \in V : A(v) \leq t < B(v)\}$$

DEF 4.21 Then $(Y_t : t \geq 0)$ tracks the total population size, and is called a *binary Yule process*.

If $Y_t = n$, then $Y_t \mapsto n+1$ when the first of the n $\text{Exp}(1)$ clocks goes off. Thus, for this process, we move $n \mapsto n+1$ at rate n , since the minimum of exponential variables is exponential with rate equal to the sum of constituent rates. (See our first discussions on continuous-time chains).

We can thus view Y_t as a continuous-time Markov chain with state space $V = \mathbb{Z}_{>0}$ and generator $Q : Q_{n,n+1} = n, Q_{n,n} = -n$, and 0 otherwise, i.e.

$$Q = \begin{bmatrix} -1 & 1 & 0 & 0 & \cdots \\ 0 & -2 & 2 & 0 & \cdots \\ 0 & 0 & -3 & 3 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

PROP 4.10 Let Y_t be a Yule process. Then

$$\frac{d}{dt} \mathbb{E}[Y_t] = \mathbb{E}[Y_t] \implies \mathbb{E}[Y_t] = e^t$$

PROOF.

$$\frac{d}{dt}\mathbb{E}[Y_t] = \frac{d}{dt}\mathbb{E}_n[\mathbb{E}[Y_t|Y_t = n]] = \sum_{n \geq 1} \frac{d}{dt}\mathbb{E}[Y_t|Y_t = n]\mathbb{P}(Y_t = n)$$

where the conditional t is fixed. Then, using the definition of the derivative,

$$\frac{d}{dt}\mathbb{E}[Y_t|Y_t = n] = \lim_{h \rightarrow 0} \frac{\mathbb{E}[Y_{t+h}|Y_t = n] - \mathbb{E}[Y_t|Y_t = n]}{h}$$

We have $\mathbb{E}[Y_t|Y_t = n] = n$. Now, if $Y_t = n$, then $Y_{t+h} = n + 1$ if $\text{Exp}(n) < h$ (i.e. we have "rung"), which occurs with probability $\approx nh + O(h^2)$. Similarly, the probability that $Y_{t+h} = n$ is $\approx 1 - nh - O(h^2)$. All totaled, then, we simplify

$$\frac{d}{dt}\mathbb{E}[Y_t|Y_t = n] = \lim_{h \rightarrow 0} \frac{(n+1)nh + n(1 - nh) - n}{h} = \lim_{h \rightarrow 0} \frac{nh}{h} = n$$

so that

$$\frac{d}{dt}\mathbb{E}[Y_t] = \sum_{n \geq 1} n\mathbb{P}(Y_t = n) = \mathbb{E}[Y_t]$$

since Y_t is a positive, integer-value variable. Solving the ODE $\mathbb{E}[Y_t]' = \mathbb{E}[Y_t]$ gives $\mathbb{E}[Y_t] = Ce^t$ with $\mathbb{E}[Y_0] = 1$, so indeed $\mathbb{E}[Y_t] = e^t$. $\square$

We can generalize [Def 4.21](#) by letting an integer random variable $B : \mathbb{P}(B = 1) = 0$ dictate the distribution of children an individual has at death. Note that we set $\mathbb{P}(B = 1) = 0$ to ensure that the continuous-time chain always changes state. In this case, we call the process a *generalized Yule process*.

DEF 4.22

The generating matrix Q for a generalized Yule process, with state space $V = \mathbb{Z}_{>0}$, has rows given by

$$\begin{bmatrix} 0 & 0 & \cdots & 0 & n\mathbb{P}(B=0) & -n & n\mathbb{P}(B=2) & n\mathbb{P}(B=3) & \cdots \\ Q_{n1} & Q_{n2} & & Q_{n(n-2)} & Q_{n(n-1)} & Q_{nn} & Q_{n(n+1)} & Q_{n(n+2)} & \end{bmatrix}$$

Birth-and-Death Processes

When $\mathbb{P}(B = 2) = p$ and $\mathbb{P}(B = 0) = 1 - p$, we specialize to

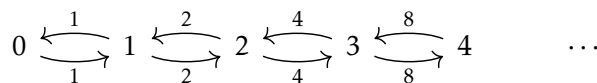
$$\begin{bmatrix} 0 & 0 & \cdots & 0 & n(1-p) & -n & np & 0 & \cdots \\ Q_{n1} & Q_{n2} & & Q_{n(n-2)} & Q_{n(n-1)} & Q_{nn} & Q_{n(n+1)} & Q_{n(n+2)} & \end{bmatrix}$$

Any such process with a tri-diagonal generating matrix is called a *birth-and-death process*. DEF 4.23

It is standard to write $\lambda_i = Q_{i,i+1}$ (called the *birth rate*) and $\mu_i = Q_{i,i-1}$ (called the *death rate*). We thus write $Q_{i,i} = -(\lambda_i + \mu_i)$. DEF 4.24

DEF 4.25

Eg. 4.7 Birth-and-death chains may be explosive. Consider an instance



where edge weights refer to the rates Q . Heuristically, the mean time spent at a state n is inversely exponential in n , i.e. $\frac{1}{2^n}$. We will transition "faster" on higher states, and so will trend upwards on the chain.

Eg. 4.8 For *non-explosive* birth-and-death chains with stationary distributions, they will be reversible by [Thm 4.3](#). Thus, the stationary distribution will satisfy

$$\pi_n \lambda_n = \pi_{n+1} \mu_{n+1} \implies \pi_{n+1} = \pi_n \frac{\lambda_n}{\mu_{n+1}}$$

by local balance ([Def 4.17](#)). We can then deduce

$$\pi_n = \pi_0 \frac{\lambda_0 \cdots \lambda_{n-1}}{\mu_1 \cdots \mu_n}$$

for the stationary distribution. To determine π_0 , we should scale by the sum of components.

Theorem 4.5 Birth-and-Death Stationarity

If a birth-and-death chain has

$$\sum_{n \geq 1} \frac{\lambda_0 \cdots \lambda_{n-1}}{\mu_1 \cdots \mu_n} < \infty$$

then the chain has a stationary distribution π , given by

$$\pi_n = \pi_0 \frac{\lambda_0 \cdots \lambda_{n-1}}{\mu_1 \cdots \mu_n}$$

where π_0 is taken to be

$$\pi_0 = \left(\sum_{n \geq 1} \frac{\lambda_0 \cdots \lambda_{n-1}}{\mu_1 \cdots \mu_n} \right)^{-1}$$

Otherwise, the chain has no stationary distribution.

Let $(X_t : t \geq 0)$ be a birth-and-death chain. Let $\tau_k := \inf\{t \geq 0 : X_t = k\}$ be the continuous variant of the hitting time for state k .

PROP 4.11 Let $(X_t : t \geq 0)$ be a birth-and-death chain with rates λ_i and μ_i . Then, for $n \geq 1$,

$$\lambda_n \mathbb{E}_n[\tau_{n+1}] = 1 + \mu_n \mathbb{E}_{n-1}[\tau_n]$$

PROOF.

We can write $\tau_{n+1} = T_1 + \tau_{n+1} - T_1$, where T_1 is the jump time. Taking expectation,

$$\mathbb{E}_n[\tau_{n+1}] = \mathbb{E}_n[T_1] + \mathbb{E}_n[\tau_{n+1} - T_1]$$

We know that $T_1 \sim \text{Exp}(\lambda_n + \mu_n)$, since we take the minimum of the birth and death

This is *not* the jump time of the chain.

Recall that $\mathbb{E}_n$ is the expectation given that $X_0 = n$

clocks. Expanding a conditional,

$$\begin{aligned}\mathbb{E}_n[\tau_{n+1}] &= \frac{1}{\lambda_n + \mu_n} + \mathbb{E}_n[\tau_{n+1} - T_1 | X_{T_1} = n+1] \mathbb{P}_n(X_{T_1} = n+1) \\ &\quad + \underbrace{\mathbb{E}_n[\tau_{n+1} - T_1 | X_{T_1} = n-1] \mathbb{P}_n(X_{T_1} = n-1)}_{\substack{\mathbb{E}_{n-1}[\tau_n] + \mathbb{E}_n[\tau_{n+1}] \\ \frac{\mu_n}{\lambda_n + \mu_n}}} \\ &= \frac{1}{\lambda_n + \mu_n} + (\mathbb{E}_{n-1}[\tau_n] + \mathbb{E}_n[\tau_{n+1}]) \frac{\mu_n}{\lambda_n + \mu_n}\end{aligned}$$

Rearranging gives the result. $\square$

As a consequence, $(X_t : t \geq 0)$ is explosive if and only if $\sum_{n \geq 0} \mathbb{E}_n[\tau_{n+1}] < \infty$.

PROP 4.12

For any continuous-time Markov chain,

PROP 4.13

$$\mathbb{E}_u[\tau_v] = \frac{1}{q_u} + \sum_{w \in V} \frac{Q_{u,w}}{q_u} \mathbb{E}_w[\tau_v]$$

Fig. 4.9 In the special case where $\lambda_n = \lambda$ and $\mu_n = \mu$, we have that

$$\pi_k = \pi_0 \left(\frac{\lambda}{\mu} \right)^k$$

If $\lambda < \mu$, then $\sum_{n \geq 0} \left(\frac{\lambda}{\mu} \right)^n = \frac{1}{1 - \frac{\lambda}{\mu}}$, so in fact $\pi_k = \frac{(\frac{\lambda}{\mu})^k}{1 - \frac{\lambda}{\mu}}$. If $\lambda \geq \mu$, no stationary distribution exists.

Investigating further, we get that

$$\mathbb{E}_n[\tau_{n+1}] = \frac{1}{\lambda} + \frac{\mu}{\lambda} \mathbb{E}_{n-1}[\tau_n]$$

Evaluating these terms explicitly, one finds that

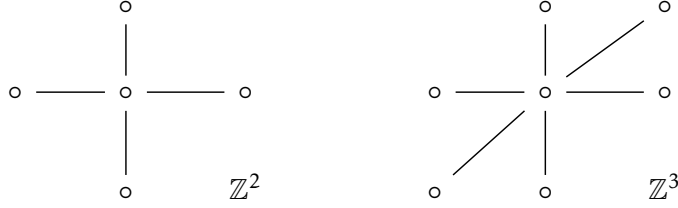
$$\mathbb{E}_n[\tau_{n+1}] = \frac{1}{\lambda} \sum_{i=0}^n \left(\frac{\mu}{\lambda} \right)^i = \begin{cases} \frac{1}{\lambda} \frac{(\frac{\mu}{\lambda})^{n+1} - 1}{\frac{\mu}{\lambda} - 1} & \lambda \neq \mu \\ \frac{1}{\lambda} (n+1) & \lambda = \mu \end{cases}$$

A FINAL EXAMPLE

This will be the closing example for this course. It does not relate to the content just seen (Yule processes, birth-and-death chains, etc.).

We consider a simple, discrete random walk on $\mathbb{Z}^d$. Previously, we used Stirling's Formula ([Example 1.11](#)) to show that a symmetric random walk on $\mathbb{Z}$ (i.e. $p = \frac{1}{2}$) is recurrent. We also use the fact that, if $(a_n)_{n \geq 0}$ is non-negative with $\sum_{n \geq 0} a_n = \infty$, and $(b_n)_{n \geq 0}$ is such

that $\frac{a_n}{b_n} \rightarrow 1$, then $\sum_{n \geq 0} b_n = \infty$. What about $\mathbb{Z}^2$ or $\mathbb{Z}^3$? Even simpler, let's consider 2 random walks on $\mathbb{Z}$ (or 3 random walks on $\mathbb{Z}$). Note that these *are* distinct questions. The possibilities from a node for the first question look like



While, for n walks on $\mathbb{Z}^n$, we always get 2^{n+1} possibilities, with differing "coordinates," i.e. $(\pm 1, \pm 1)$ for 2 walks, and $(\pm 1, \pm 1, \pm 1)$ for 3 walks. These processes:

$$((X_n, Y_n) : n \geq 0) \quad ((X_n, Y_n, Z_n) : n \geq 0)$$

are both Markov chains on the states spaces $\mathbb{Z}^2$ and $\mathbb{Z}^3$, respectively.

PROP 4.14 There are two communication classes of (X_n, Y_n) , and they are both recurrent.

PROOF.

It suffices to prove that

$$\sum_{n \geq 0} \mathbb{P}((X_{2n}, Y_{2n}) = (0, 0) | (X_0, Y_0) = (0, 0))$$

diverges. We can express the summands as

$$\begin{aligned} & \mathbb{P}(X_{2n} = 0 | (X_0, Y_0) = (0, 0)) \mathbb{P}(Y_{2n} = 0 | (X_0, Y_0) = (0, 0)) \\ &= \mathbb{P}(X_{2n} = 0 | X_0 = 0) \mathbb{P}(Y_{2n} = 0 | Y_0 = 0) \\ &\approx \frac{1}{\sqrt{\pi n}} \cdot \frac{1}{\sqrt{\pi n}} = \frac{1}{\pi n} \end{aligned}$$

by Sterling's Formula. A sum of these terms diverges. □

PROP 4.15 There are 4 communication classes of (X_n, Y_n, Z_n) , and all are transient.

PROOF.

Independence gives that, similar to above,

$$\sum_{n \geq 0} \mathbb{P}(X_{2n} = Y_{2n} = Z_{2n} | X_0 = Y_0 = Z_0 = 0) \approx \sum_{n \geq 0} \left(\frac{1}{\sqrt{q n}} \right)^3 < \infty$$

□

Theorem 4.6 Polyá

Symmetric simple random walks on $\mathbb{Z}^2$ are recurrent. On $\mathbb{Z}^d$ for $d \geq 3$, however, this is transient.

DEFINITIONS

- absorbing 1.26
- accessible 1.14
- accessible 4.12
- aperiodic 1.19
- arrival time 3.3

- binary Yule process 4.21
- birth rate 4.24
- birth-and-death process 4.23
- branching process 2.1

- canonical decomposition 1.24
- closed 1.25
- communicate 1.15
- communication class 1.16
- conditional density 1.1
- continuous time-homogenous 4.4
- continuous-time Markov chain 4.3
- counting process 3.2

- death rate 4.25
- directed transition graph 4.20

- embedded chain 4.8
- explosive 4.7
- exponential alarm clock 4.10
- extinct 2.3

- forward and backward equations 4.11

- generalized Yule process 4.22
- generator matrix 4.2
- global balance equation 4.16

- hitting time 1.7
- holding time 4.9
- homogeneous Poisson process 3.1

- initial distribution 1.9
- interarrival times 3.4
- irreducible 1.17
- irreducible 4.13
- jump times 4.5

- law of large numbers 1.10
- limiting 4.15
- limiting distribution 1.12
- local balance equation 4.17
- local time 1.23

- Markov chain 1.2
- Markov property 1.4
- Metropolis-Hastings 1.30
- moment generating function 2.4
- Monte-Carlo algorithm 1.29

- non-explosive 4.6

- period 1.18
- Poisson process 3.5

- recurrent 1.21
- regular 1.13
- return time 1.20
- reversible 1.28

- state space 1.3
- stationary 1.11
- stationary 4.14
- stochastic matrix 1.8
- survives 2.2

- time reversal 1.27
- time-homogeneous 1.5
- time-reversible 4.18
- transient 1.22
- transition function 4.1
- transition matrix 1.6
- type 3.6

- undirected transition graph 4.19

THEOREMS

Birth-and-Death Stationarity [4.5](#)

Chapman-Kolmogorov [1.2](#)

Chapman-Kolmogorov for CTMC [4.2](#)

Characterization of Regular Markov Chains [1.5](#)

Continuous-Time Markov Characterization [4.1](#)

Criterion for Continuous-Time Reversibility [4.3](#)

Existence Theorem [1.6](#)

Fundamental Theorem of Branching Processes [2.1](#)

Fundamental Theorem of Continuous-Time Markov Chains [4.4](#)

Fundamental Theorem of Markov Chains [1.3](#)

Hitting States in (Transient, Absorbing) Chains [1.7](#)

Markov Property [1.1](#)

Norris' Theorem [1.8](#)

Polyá [4.6](#)

Postage Stamp Lemma [1.4](#)

Reversible Chains are Random Weighted Walks [1.9](#)